1. Details of parameter

List of Parameters

Related Control Mode				Parametr No.		Title	Detail page
P	S	T	F	Class	No.		
P	S	T	F	[Class 0] Basic setting	00	Rotational direction setup	4-4
P	S	T	F		01	Control mode setup	
P	S	T	F		02	Real-time auto-gain tuning setup	4-5
P	S	T	F		03	Selection of machine stiffness at real-time auto-gain tuning	
P	S	T	F		04	Inertia ratio	4-6
P	S	T	F		05	Selection of command pulse input	
P	S	T	F		06	Command pulse rotational direction setup	4-7
P	S	T	F		07	Command pulse input mode setup	
P	S	T	F		08	Command pulse counts per one motor revolution	4-8
P	S	T	F		09	1st numerator of electronic gear	
P	S	T	F		10	Denominator of electronic gear	
P	S	T	F		11	Output pulse counts per one motor revolution	4-9
P	S	T	F		12	Reversal of pulse output logic	4-10
P	S	T	F		13	1st torque limit	
P	S	T	F		14	Position deviation excess setup	
P	S	T	F		15	Absolute encoder setup	
P	S	T	F		16	External regenerative resistor setup	4-11
P	S	T	F		17	Load factor of external regenerative resistor selection	
P	S	T	F	[Class 1] Gain adjustment	00	1st gain of position loop	4-12
P	S	T	F		01	1st gain of velocity loop	
P	S	T	F		02	1st time constant of velocity loop integration	
P	S	T	F		03	1st filter of speed detection	
P	S	T	F		04	1st time constant of torque filter	
P	S	T	F		05	2nd gain of position loop	4-13
P	S	T	F		06	2nd gain of velocity loop	
P	S	T	F		07	2nd time constant of velocity loop integration	
P	S	T	F		08	2nd filter of speed detection	
P	S	T	F		09	2nd time constant of torque filter	
P	S	T	F		10	Velocity feed forward gain	
P	S	T	F		11	Velocity feed forward filter	
P	S	T	F		12	Torque feed forward gain	4-14
P	S	T	F		13	Torque feed forward filter	
P	S	T	F		14	2nd gain setup	
P	S	T	F		15	Mode of position control switching	4-15
P	S	T	F		16	Delay time of position control switching	4-16
P	S	T	F		17	Level of position control switching	
P	S	T	F		18	Hysteresis at position control switching	
P	S	T	F		19	Position gain switching time	
P	S	T	F		20	Mode of velocity control switching	4-17
P	S	T	F		21	Delay time of velocity control switching	
P	S	T	F		22	Level of velocity control switching	
P	S	T	F		23	Hysteresis at velocity control switching	
P	S	T	F		24	Mode of torque control switching	4-18
P	S	T	F		25	Delay time of torque control switching	
P	S	T	F		26	Level of torque control switching	
P	S	T	F		27	Hysteresis at torque control switching	
P	S	T	F	[Class 2]	00	Adaptive filter mode setup	4-19
P	S	T	F		01	1st notch frequency	
P	S	T	F		02	1st notch width selection	
P	S	T	F		03	1st notch depth selection	
P	S	T	F		04	2nd notch frequency	

Related Control Mode				Parametr No.		Title	Detail page
P	S	T	F	Class	No.		
P	S	T	F	[Class 2] Damping control	05	2nd notch width selection	4-20
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P	S	T	F		07	3rd notch frequency	
P	S	T	F		08	3rd notch width selection	
P	S	T	F		09	3rd notch depth selection	
P	S	T	F		10	4th notch frequency	
P	S	T	F		11	4th notch width selection	4-21
P	S	T	F		12	4th notch depth selection	
P	S	T	F		13	Selection of damping filter switching	
P	S	T	F		14	1st damping frequency	4-22
P	S	T	F		15	1st damping filter setup	
P	S	T	F		16	2nd damping frequency	
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P	S	T	F		19	3rd damping filter setup	
P	S	T	F		20	4th damping frequency	
P	S	T	F		21	4th damping filter setup	
P	S	T	F		22	Positional command smoothing filter	4-23
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P	S	T	F	[Class 3] Verocity/ Torque/ Full-closed control	00	Speed setup, Internal/External switching	4-24
P	S	T	F		01	Speed command rotational direction selection	
P	S	T	F		02	Input gain of speed command	4-25
P	S	T	F		03	Reversal of speed command input	
P	S	T	F		04	1st speed of speed setup	4-26
P	S	T	F		05	2nd speed of speed setup	
P	S	T	F		06	3rd speed of speed setup	
P	S	T	F		07	4th speed of speed setup	
P	S	T	F		08	5th speed of speed setup	
P	S	T	F		09	6th speed of speed setup	
P	S	T	F		10	7th speed of speed setup	
P	S	T	F		11	8th speed of speed setup	
P	S	T	F		12	Acceleration time setup	
P	S	T	F		13	Deceleration time setup	
P	S	T	F		14	Sigmoid acceleration/ deceleration time setup	4-27
P	S	T	F		15	Speed zero-clamp function selection	
P	S	T	F		16	Speed zero clamp level	
P	S	T	F		17	Selection of torque command	4-28
P	S	T	F		18	Torque command direction selection	
P	S	T	F		19	Input gain of torque command	
P	S	T	F		20	Input reversal of torque command	
P	S	T	F		21	Speed limit value 1	4-29
P	S	T	F		22	Speed limit value 2	
P	S	T	F		23	External scale selection	
P	S	T	F		24	Numerator of external scale division	4-30
P	S	T	F		25	Denominator of external scale division	
P	S	T	F		26	Reversal of direction of external scale	
P	S	T	F		27	External scale Z phase disconnection detection disable	
P	S	T	F		28	Hybrid deviation excess setup	4-31
P	S	T	F		29	Hybrid deviation clear setup	

Note • Definition of symbols under “Related mode” - P: position control, S: velocity control, T: torque control, F: full closed control

Related Control Mode				Parametr No.		Title	Detail page
P	S	T	F	Class	No.		
P	S	T	F	[Class 4] I/F monitor setting	00	SI1 input selection	4-32
P	S	T	F		01	SI2 input selection	4-33
P	S	T	F		02	SI3 input selection	
P	S	T	F		03	SI4 input selection	
P	S	T	F		04	SI5 input selection	
P	S	T	F		05	SI6 input selection	
P	S	T	F		06	SI7 input selection	
P	S	T	F		07	SI8 input selection	
P	S	T	F		08	SI9 input selection	
P	S	T	F		09	SI10 input selection	
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P	S	T	F		11	SO2 output selection	
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P	S	T	F		15	SO6 output selection	
P	S	T	F		16	Type of analog monitor 1	4-35
P	S	T	F		17	Analog monitor 1 output gain	
P	S	T	F		18	Type of analog monitor 2	
P	S	T	F		19	Analog monitor 2 output gain	
P	S	T	F		20	Type of digital monitor	
P	S	T	F		21	Analog monitor output setup	4-37
P	S	T	F		22	Analog input 1 (AI1) offset setup	
P	S	T	F		23	Analog input 1 (AI1) filter	
P	S	T	F		24	Analog input 1 (AI1) overvoltage setup	
P	S	T	F		25	Analog input 2 (AI2) offset setup	
P	S	T	F		26	Analog input 2 (AI2) filter	
P	S	T	F		27	Analog input 2(AI2) overvoltage setup	
P	S	T	F		28	Analog input 3 (AI3) offset setup	4-38
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P	S	T	F		30	Analog input 3 (AI3) overvoltage setup	
P	S	T	F		31	Positioning complete (In-position) range	
P	S	T	F		32	Positioning complete (In-position) output setup	
P	S	T	F		33	INP hold time	4-39
P	S	T	F		34	Zero-speed	
P	S	T	F		35	Speed coincidence range	4-40
P	S	T	F		36	At-speed (Speed arrival)	
P	S	T	F		37	Mechanical brake action at stalling setup	4-41
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P	S	T	F		39	Brake release speed setup	
P	S	T	F		40	Selection of alarm output 1	4-42
P	S	T	F		41	Selection of alarm output 2	
P	S	T	F		42	2nd Positioning complete (In-position) range	
P	S	T	F	[Class 5] Enhancing setting	00	2nd numerator of electronic gear	4-43
P	S	T	F		01	3rd numerator of electronic gear	
P	S	T	F		02	4th numerator of electronic gear	
P	S	T	F		03	Denominator of pulse output division	
P	S	T	F		04	Over-travel inhibit input setup	
P	S	T	F		05	Sequence at over-travel inhibit	
P	S	T	F		06	Sequence at Servo-Off	4-44
P	S	T	F		07	Sequence at main power OFF	
P	S	T	F		08	LV trip selection at main power OFF	4-45
P	S	T	F		09	Detection time of main power off	
P	S	T	F		10	Sequence at alarm	
P	S	T	F		11	Torque setup for emergency stop	4-46
P	S	T	F		12	Over-load level setup	

Related Control Mode				Parametr No.		Title	Detail page
P	S	T	F	Class	No.		
P	S	T	F	[Class 5] Enhancing setting	13	Over-speed level setup	4-46
P	S	T	F		14	Motor working range setup	
P	S	T	F		15	I/F reading filter	
P	S	T	F		16	Alarm clear input setup	
P	S	T	F		17	Counter clear input mode	4-47
P	S	T	F		18	Invalidation of command pulse inhibit input	
P	S	T	F		19	Command pulse inhibit input reading setup	
P	S	T	F		20	Position setup unit select	
P	S	T	F		21	Selection of torque limit	4-48
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P	S	T	F		23	Torque limit switching setup 1	
P	S	T	F		24	Torque limit switching setup 2	
P	S	T	F		25	External input positive direction torque limit	
P	S	T	F		26	External input negative direction torque limit	4-49
P	S	T	F		27	Input gain of analog torque limit	
P	S	T	F		28	LED initial status	
P	S	T	F		29	RS232 baud rate setup	
P	S	T	F		30	RS485 baud rate setup	4-50
P	S	T	F		31	Axis address	
P	S	T	F		32	Command pulse input maximum setup	
P	S	T	F		33	Pulse regenerative output limit setup	
P	S	T	F		34	For manufacturer's use	
P	S	T	F		35	Front panel lock setup	
P	S	T	F	[Class 6] Special setting	00	Analog torque feed forward conversion gain	4-51
P	S	T	F		02	Velocity deviation excess setup	
P	S	T	F		04	JOG trial run command speed	
P	S	T	F		05	Position 3rd gain valid time	
P	S	T	F		06	Position 3rd gain scale factor	
P	S	T	F		07	Torque command additional value	
P	S	T	F		08	Positive direction torque compensation value	4-52
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P	S	T	F		10	Function expansion setup	
P	S	T	F		11	Current response setup	
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P	S	T	F		15	2nd over-speed level setup	
P	S	T	F		17	Front panel parameter writing selection	
P	S	T	F		18	Power-up wait time	
P	S	T	F		19	Encoder Z phase setup	
P	S	T	F		20	Z-phase setup of external scale	
P	S	T	F		21	Serial absolute external scale Z phase setup	4-54
P	S	T	F		22	A, B phase external scale pulse output method selection	
P	S	T	F		23	Disturbance torque compensating gain	
P	S	T	F		24	Disturbance observer filter	
P	S	T	F		27	Alarm latch time selection	
P	S	T	F		31	Real time auto tuning estimation speed	4-55
P	S	T	F		32	Real time auto tuning custom setup	
P	S	T	F		34	Hybrid vibration suppression gain	4-57
P	S	T	F		35	Hybrid vibration suppression filter	
P	S	T	F		37	Oscillation detection level	
P	S	T	F		38	Alarm mask setup	
P	S	T	F		39	For manufacturer's use	

1. Details of parameter

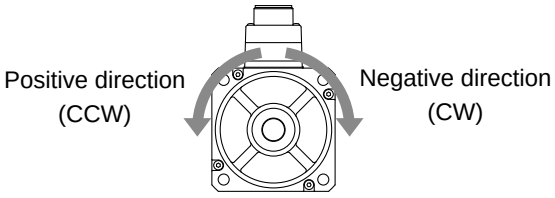
[Class 0] Basic setting

Pr0.00 *	Title	Rotational direction setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	1			

Setup the relationship between the direction of command and direction of motor rotation.

0: Motor turns CW in response to positive direction command (CW when viewed from load side shaft end)

1: Motor turns CCW in response to positive direction command (CCW when viewed from load side shaft end)



Default

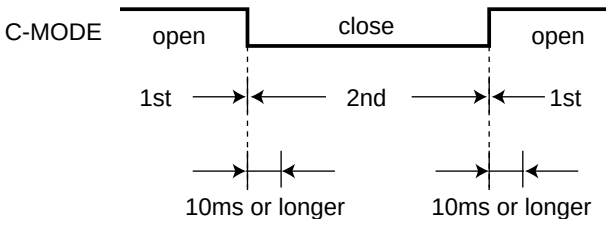
Setup value	Command direction	Motor rotational direction	Positive direction drive inhibit input	Negative direction drive inhibit input
0	Positive direction	CW	Valid	—
	Negative direction	CCW	—	Valid
1	Positive direction	CCW	Valid	—
	Negative direction	CW	—	Valid

Pr0.01 *	Title	Control mode setup			Related control mode	P	S	T	F
	Range	0 to 6	Unit	—	Default	0			

You can set up the control mode to be used.

Setup value	Content	
	1st mode	2st mode
0	Position	—
1	Velocity	—
2	Torque	—
3 *1	Position	Velocity
4 *1	Position	Torque
5 *1	Velocity	Torque
6	Full-closed	—

*1) When you set up the combination mode of 3, 4 or 5, you can select either the 1st or the 2nd with control mode switching input (C-MODE).
When C-MODE is open, the 1st mode will be selected.
When C-MODE is shorted, the 2nd mode will be selected.
Don't enter commands 10ms before/after switching.



The waveform above shows when logical setting of C-MODE input is a-contact. When b-contact is used, open and short is reversed.

Note • **For parameters which No. have a suffix of " * ",** changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr0.02	Title	Real-time auto-gain tuning setup			Related control mode	P	S	T	F
	Range	0 to 6	Unit	—	Default	1			

You can set up the action mode of the real-time auto-gain tuning.

Setup value	Mode	Varying degree of load inertia in motion
0	Invalid	Real-time auto-gain tuning function is disabled.
1	Standard	Basic mode. Do not use unbalanced load, friction compensation or gain switching.
2	Positioning *1	Main application is positioning. It is recommended to use this mode on equipment without unbalanced horizontal axis, ball screw driving equipment with low friction, etc.
3	Vertical axis *2	With additional features to the positioning mode - use this mode to positively and effectively compensate for unbalanced load to the vertical axis or minimize variations in setting time.
4	Friction compensation *3	With additional features to the vertical axis mode - use this mode to positively and effectively reduce positioning setting time when the belt driving axis has high friction.
5	Load characteristic measurement	Estimate the load characteristics without changing current parameter setting. This mode requires use of the setup support software.
6	Customize *4	Functions of real-time auto-gain tuning can be customized to meet the requirements of the specific application by combining desired functions according to the Pr6.32 “Real-time auto-gain tuning custom setting”.

*1 Velocity and torque controls are the same as in the standard mode.
*2 Torque control is the same as in the standard mode.
*3 Velocity control is the same as in the vertical axis mode. Torque control is the same as in the standard mode.
*4 Certain function(s) is not available in a specific control mode. Refer to description in Pr6.32.

Pr0.03	Title	Selection of machine stiffness at real-time auto-gain tuning			Related control mode	P	S	T	F
	Range	0 to 31	Unit	—	Default	A,B,C-frame		13	
						D,E,F-frame		11	

You can set up the response while the real-time auto-gain tuning is valid.

low ← machine stiffness → high
low ← servo gain → high
0, 1 - - - - - - - 11 - - 13 - - - - - - - - - - - - 30, 31
low ← response → high

Caution Higher the setup value, higher the velocity response and servo stiffness will be obtained. However, when increasing the value, check the resulting operation to avoid oscillation or vibration.

Pr0.04	Title	Inertia ratio			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	%	Default	250 *			

Set 1st inertia ratio.
You can set up the ratio of the load inertia against the rotor (of the motor) inertia.

Pr0.04 = (load inertia/ rotor inertia) × 100 [%]

The inertia ratio will be estimated at all time while the real-time auto-gain tuning is valid, and its result will be saved to EEPROM every 30 min.

Caution

If the inertia ratio is correctly set, the setup unit of Pr1.01 and Pr1.06 becomes (Hz). When the inertia ratio of Pr0.04 is larger than the actual, the setup unit of the velocity loop gain becomes larger, and when the inertia ratio of Pr0.04 is smaller than the actual, the setup unit of the velocity loop gain becomes smaller.

Pr0.05 *	Title	Selection of command pulse input			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

You can select either the photo-coupler input or the exclusive input for line driver as the command pulse input.

Setup value	Content
0	Photo-coupler input (PULS1, PULS2, SIGN1, SIGN2)
1	Exclusive input for line driver (PULSH1, PULSH2, SIGNH1, SIGNH2)

Note

- Parameters which default values have a suffix of " * " will be automatically set up during real time auto-gain tuning. When you change manually, invalidate the real-time auto-gain tuning first then set, referring to P.5-8, "Release of Automatic Gain Adjusting Function" of Adjustment.

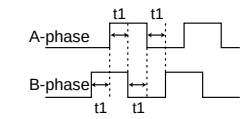
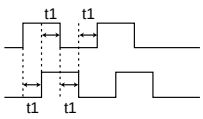
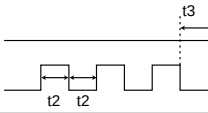
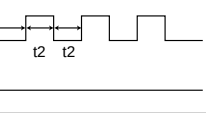
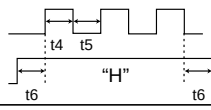
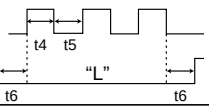
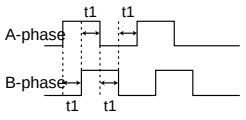
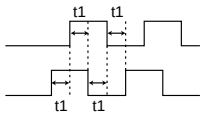
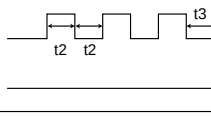
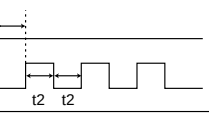
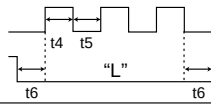
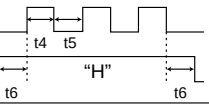
Pr0.06 *	Title	Command pulse rotational direction setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Pr0.07 *	Title	Command pulse input mode setup			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	1			

You can set up the rotational direction against the command pulse input, and the command pulse input format.

The table below shows combinations of Pr0.06 Command pulse rotational direction setup and Pr0.07 Command pulse input mode setup.
Pulses are counted at edges indicated by the arrows as shown in the table.

• Input format command pulse

Pr0.06 setup value (Command pulse rotational direction setup)	Pr0.07 setup value (Command pulse input mode setup)	Command pulse format	Signal title	Positive direction command	Negative direction command
0	0 or 2	90˚ phase difference 2-phase pulse (A + B-phase)	PULS SIGN	 B-phase advances to A by 90˚.	 B-phase delays from A by 90˚.
	1	Positive direction pulse train + Negative direction pulse train	PULS SIGN		
	3	pulse train + Signal	PULS SIGN		
1	0 or 2	90˚ phase difference 2-phase pulse (A + B-phase)	PULS SIGN	 B-phase delays from A by 90˚.	 B-phase advances to A by 90˚.
	1	Positive direction pulse train + Negative direction pulse train	PULS SIGN		
	3	pulse train + Signal	PULS SIGN		

• Permissible max. input frequency, and min. necessary time width of command pulse input signal.

Input I/F of PULS/SIGN signal		Permissible max. input frequency	Min. necessary time width (µs)					
			t_1	t_2	t_3	t_4	t_5	t_6
Pulse train interface exclusive to line driver		4Mpps	0.25	0.125	0.125	0.125	0.125	0.125
Pulse train interface	Line driver interface	500kpps	2	1	1	1	1	1
	Open collector interface	200kpps	5	2.5	2.5	2.5	2.5	2.5

Make the rising/falling time of the command pulse input signal to 0.1µs or smaller.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr0.08 *	Title	Command pulse counts per one motor revolution			Related control mode	P S T F
	Range	0 to 1048576	Unit	pulse	Default	10000

Set the command pulses that causes single turn of the motor shaft.
When this setting is 0, Pr0.09 1st numerator of electronic gear and Pr0.10 Denominator of electronic gear become valid.

Pr0.09	Title	1st numerator of electronic gear			Related control mode	P S T F
	Range	0 to 2^{30}	Unit	—	Default	0

Set the numerator of division/multiplication operation made according to the command pulse input.
This setup is enabled when Pr0.08 command pulse counts per one motor revolution = 0.
When the setup value is 0, the numerator is replaced by the encoder resolution.

Pr0.10	Title	Denominator of electronic gear			Related control mode	P S T F
	Range	1 to 2^{30}	Unit	—	Default	10000

Set the Denominator of division/multiplication operation made according to the command pulse input.
This setup is enabled when Pr0.08 command pulse counts per one motor revolution = 0.

Pr0.08	Pr0.09	Pr0.10	Command division/multiplication operation
1 to 1048576	— (Not effect)	— (Not effect)	Command pulse input → Encoder resolution / [Pr0.08 setup value] → Positional command *** Regardless of setup of Pr0.09 and Pr0.10, this operation is processed according to setup value of Pr0.08.**
0	**0**	**0 to 1073741824**	Command pulse input → Encoder resolution / [Pr0.10 setup value] → Positional command *** When both Pr0.09 and Pr0.09 are set to 0, this operation is processed according to setup value of Pr0.10.**
	1 to 1073741824	**1 to 1073741824**	Command pulse input → [Pr0.09 setting] / [Pr0.10 setting] → Positional command *** When setup value of Pr0.08 is 0, and Pr0.09≠0, this operation is processed according to setup value of Pr0.09 and Pr0.10.**

Related page • P4-62 “Setup of Motor Rotational Speed and Input Pulse Frequency”

Pr0.11 *	Title	Output pulse counts per one motor revolution			Related control mode	P	S	T	F
	Range	1 to 262144	Unit	P/r	Default	2500			

You can set up the output pulse counts per one motor revolution for each OA and OB with the Pr0.11 setup. Therefore the pulse output resolution after quadruple can be obtained from the formula below.
The pulse output resolution per one revolution = Pr0.11 setup value × 4

Pr5.03 *	Title	Denominator of pulse output division			Related control mode	P	S	T	F
	Range	0 to 262144	Unit	—	Default	0			

For an application where the number of output pulses per one motor revolution is not an integer, set this parameter to a value other than 0; and the dividing ratio can be set by using Pr0.11 as the numerator and Pr5.03 as the denominator.
Therefore the pulse output resolution after quadruple can be obtained from the formula below.

Pulse output resolution per one revolution
= (Pr0.11 setup value / Pr5.03 setup value) × Encoder resolution

<Combination of Pr0.11 Output pulse counts per one motor revolution and Pr5.03 Denominator of pulse output division>

<table>
<tr><th>Pr0.11</th><th>Pr5.03</th><th>Pulse reproducing process</th></tr>
<tr><td rowspan="2">1 to 262144</td><td>0</td><td>When the output source is encoder
Encoder pulse [pulse] → [Pr0.11 setup value]×4 / Encoder resolution → Output pulse [pulse]
* When Pr5.03 = 0, the above process is made according to Pr0.11 setup value.
The number of pulses of reproduced pulse output OA and OB are the number of pulses set in Pr0.11. The resolution of pulse output per one revolution is equal to or less the encoder resolution.
When the output source is external scale
External scale pulse [pulse] → 1 / 1 → Output pulse [pulse]
* Division ratio is 1:1.</td></tr>
<tr><td>1 to 262144</td><td>Encoder FB pulse or external scale pulse [pulse] → [Pr0.11 setup value] / [Pr5.03 setup value] → Output pulse [pulse]
* If Pr5.03 is not equal to 0, then the above process is performed based on setup value of Pr0.11 and Pr5.03. This feature can be used for application where the number of reproduced pulses (OA, OB) per one motor revolution is not an integer.
Note that when the number of pulses per one motor revolution is not an integer, Z-phase output is not synchronized with that of A-phase, reducing pulse width. The pulse output resolution per one revolution cannot become higher than the encoder resolution.</td></tr>
</table>

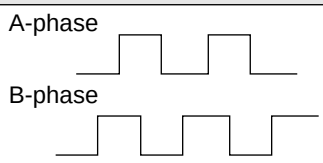
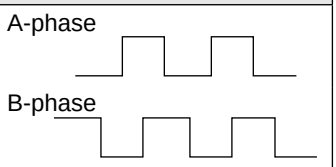
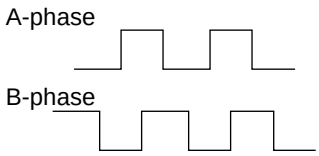
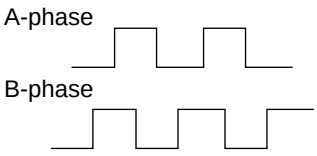
Note
- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page
- P.3-30... "Inputs and outputs on connector X4"

Pr0.12 *	Title	Reversal of pulse output logic			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

You can set up the B-phase logic and the output source of the pulse output. With this parameter, you can reverse the phase relation between the A-phase pulse and the B-phase pulse by reversing the B-phase logic. Encoder or feedback scale can be selected as the output source for full-closed control. The encoder is selected as the source if not for full-closed control.

<Reversal of pulse output logic>

Pr0.12	B-phase logic	Output source	Positive direction rotation	Negative direction rotation
0	Non-reversal	Encoder		
2		Feedback scale		
1	Reversal	Encoder		
3		Feedback scale		

Caution Setup value 2 and 3 are valid only for full-closed control. Setting must be 0 or 1 if not for full-closed control.

Pr0.13	Title	1st torque limit			Related control mode	P	S	T	F
	Range	0 to 500	Unit	%	Default	500			

You can set up the limit value of the motor output torque.

Note For details of torque limit value, refer to P.2-49.

Pr0.14	Title	Position deviation excess setup			Related control mode	P	S	T	F
	Range	0 to 2^{27}	Unit	Command unit	Default	100000			

- Set excess range of positional deviation by the command unit (default).
- Setup unit can be changed to encoder unit through Pr5.20 (position setup unit selection). If the unit is changed, set up with the encoder pulse counts at the position control and with the feedback scale pulse counts at the full-closed control.
- Err24.0 (Error detection of position deviation excess) becomes invalid when you set up this to 0.

Pr0.15 *	Title	Absolute encoder setup			Related control mode	P	S	T	F
	Range	0 to 2	Unit	—	Default	1			

You can set up the using method of 17-bit absolute encoder.

Setup value	Function
0	Use as an absolute encoder.
1	Use as an incremental encoder.
2	Use as an absolute encoder, but ignore the multi-turn counter over.

Caution This parameter will be invalidated when 5-wire, 20-bit incremental encoder is used.

Pr0.16 *	Title	External regenerative resistor setup			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	A,B-frame		3	
						C to F-frame		0	

With this parameter, you can select either to use the built-in regenerative resistor of the driver, or to separate this built-in regenerative resistor and externally install the regenerative resistor (between B1 and B2 of Connector XB in case of A to D-frame, between B1 and B2 of terminal block in case of E, F-frame).

Setup value	Regenerative resistor to be used	Function
0 (C to F-frame)	Built-in resistor	Regenerative processing circuit will be activated and regenerative resistor overload protection will be triggered according to the built-in resistor (approx. 1% duty).
1	External resistor	The driver trips due to regenerative overload protection (Err18.0), when regenerative processing circuit is activated and its active ratio exceeds 10%.
2	External resistor	Regenerative processing circuit is activated, but no regenerative over-load protection is triggered.
3 (A, B-frame)	No resistor	Both regenerative processing circuit and regenerative protection are not activated, and built-in capacitor handles all regenerative power.

Remarks Install an external protection such as thermal fuse when you use the external regenerative resistor.
Otherwise, the regenerative resistor might be heated up abnormally and result in burnout, regardless of validation or invalidation of regenerative over-load protection.

Caution When you use the built-in regenerative resistor, never to set up other value than 0. Don't touch the external regenerative resistor.
External regenerative resistor gets very hot, and might cause burning.

Pr0.17 *	Title	Load factor of external regenerative resistor selection			Related control mode	P	S	T	F
	Range	0 to 4	Unit	—	Default	0			

When selecting the external regenerative resistor (Pr0.16 = 1, 2), select the computing method of load factor of regenerative resistor.

Setup value	Function
0	Regenerative load factor is 100% when duty factor of external regenerative resistor is 10%.
1 to 4	For manufacturer's use (do not setup)

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.2-8 "System Configuration and Wiring" • P.3-30... "Inputs and outputs on connector X4"

1. Details of parameter

[Class 1] Gain adjustment

Pr1.00	Title	1st gain of position loop			Related control mode	P	S	T	F
	Range	0 to 30000	Unit	0.1/s	Default	A,B,C-frame	480		
						D,E,F-frame	320		

You can determine the response of the positional control system.
Higher the gain of position loop you set, faster the positioning time you can obtain.
Note that too high setup may cause oscillation.

Pr1.01	Title	1st gain of velocity loop			Related control mode	P	S	T	F
	Range	1 to 32767	Unit	0.1Hz	Default	A,B,C-frame	270		
						D,E,F-frame	180		

You can determine the response of the velocity loop.
In order to increase the response of overall servo system by setting high position loop gain, you need higher setup of this velocity loop gain as well. However, too high setup may cause oscillation.

Caution When the inertia ratio of Pr0.04 is set correctly, the setup unit of Pr1.01 becomes (Hz).

Pr1.02	Title	1st time constant of velocity loop integration			Related control mode	P	S	T	F
	Range	1 to 10000	Unit	0.1ms	Default	A,B,C-frame	210		
						D,E,F-frame	310		

You can set up the integration time constant of velocity loop.
Smaller the setup, faster you can dog-in deviation at stall to 0.
The integration will be maintained by setting to "9999".
The integration effect will be lost by setting to "10000".

Pr1.03	Title	1st filter of speed detection			Related control mode	P	S	T	F
	Range	0 to 5	Unit	—	Default	0			

You can set up the time constant of the low pass filter (LPF) after the speed detection, in 6 steps.
Higher the setup, larger the time constant you can obtain so that you can decrease the motor noise, however, response becomes slow. Use with a default value of 0 in normal operation.

Pr1.04	Title	1st time constant of torque filter			Related control mode	P	S	T	F
	Range	0 to 2500	Unit	0.01ms	Default	A,B,C-frame	84		
						D,E,F-frame	126		

You can set up the time constant of the 1st delay filter inserted in the torque command portion. You might expect suppression of oscillation caused by distortion resonance.

Caution
- To Panasonic MINAS users: A4 and higher series
  CAUTION: Parameter settings shown in this manual may differ from those applied to your product (s).

Note
- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page
- P.3-30... "Inputs and outputs on connector X4"

Pr1.05	Title	2nd gain of position loop			Related control mode	P S T F
	Range	0 to 30000	Unit	0.1/s	Default	A,B,C-frame 570 D,E,F-frame 380

Pr1.06	Title	2nd gain of velocity loop			Related control mode	P S T F
	Range	1 to 32767	Unit	0.1Hz	Default	A,B,C-frame 270 D,E,F-frame 180

Pr1.07	Title	2nd time constant of velocity loop integration			Related control mode	P S T F
	Range	1 to 10000	Unit	0.1ms	Default	10000

Pr1.08	Title	2nd filter of speed detection			Related control mode	P S T F
	Range	0 to 5	Unit	—	Default	0

Pr1.09	Title	2nd time constant of torque filter			Related control mode	P S T F
	Range	0 to 2500	Unit	0.01ms	Default	A,B,C-frame 84 D,E,F-frame 126

Position loop, velocity loop, speed detection filter and torque command filter have their 2 pairs of gain or time constant (1st and 2nd).

Related page For details of switching the 1st and the 2nd gain or the time constant, refer to P.5-17 "Gain Switching Function" of Adjustment.
The function and the content of each parameter is as same as that of the 1st gain and time constant.

Pr1.10	Title	Velocity feed forward gain			Related control mode	P S T F
	Range	0 to 1000	Unit	0.10%	Default	300

Multiply the velocity control command calculated according to the internal positional command by the ratio of this parameter and add the result to the speed command resulting from the positional control process.

Pr1.11	Title	Velocity feed forward filter			Related control mode	P S T F
	Range	0 to 6400	Unit	0.01ms	Default	50

Set the time constant of 1st delay filter which affects the input of velocity feed forward.

<Usage example of velocity feed forward>

The velocity feed forward will become effective as the velocity feed forward gain is gradually increased with the velocity feed forward filter set at approx. 50 (0.5 ms). The positional deviation during operation at a constant velocity is reduced as shown in the equation below in proportion to the value of velocity feed forward gain.

Positional deviation [unit of command] = command speed [unit of command/s] / positional loop gain [1/s] × (100 - velocity feed forward gain [%]) / 100

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr1.12	Title	Torque feed forward gain			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1%	Default	0			

- Multiply the torque command calculated according to the velocity control command by the ratio of this parameter and add the result to the torque command resulting from the velocity control process.
- Positional deviation at a constant acceleration/deceleration can be minimized close to 0 by increasing the torque forward gain. This means that positional deviation can be maintained at near 0 over entire operation range while driving in trapezoidal speed pattern under ideal condition where disturbance torque is not active.

Pr1.13	Title	Torque feed forward filter			Related control mode	P	S	T	F
	Range	0 to 6400	Unit	0.01ms	Default	0			

- Set up the time constant of 1st delay filter which affects the input of torque feed forward.
- The torque feed forward will become effective as the torque feed forward gain is gradually increased with the torque feed forward filter is set at approx. 50 (0.5 ms).

<Usage example of torque feed forward>

- To use the torque feed forward, correctly set the inertia ratio.
  Use the value that was determined at the start of the real time auto tuning, or set the inertia ratio that can be calculated from the machine specification to Pr0.04 Inertia ratio.
- The torque feed forward will become effective as the torque feed forward gain is gradually increased with the torque feed forward filter is set at approx. 50 (0.5 ms).
- Positional deviation at a constant acceleration/deceleration can be minimized close to 0 by increasing the torque forward gain. This means that positional deviation can be maintained at near 0 over entire operation range while driving in trapezoidal speed pattern under ideal condition where disturbance torque is not active .

Caution
Zero positional deviation is impossible in actual situation because of disturbance torque.
As with the velocity feed forward, large torque feed forward filter time constant decreases the operating noise but increases positional deviation at acceleration change point.

Pr1.14	Title	2nd gain setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	1			

Arrange this parameter when performing optimum adjustment by using the gain switching function.

Setup value	Gain selection/switching
0	1st gain is fixed at a value. By using the gain switching input (GAIN), change the velocity loop operation from PI to P. GAIN input photo-coupler OFF → PI operation GAIN input photo-coupler ON → P operation * The above description applies when the logical setting of GAIN input is a-contact. ON/OFF of photo-coupler is reversed when b-contact.
1	Enable gain switching of 1st gain (Pr1.00-Pr1.04) and 2nd gain (Pr1.05-Pr1.09).

Related page
For switching condition of the 1st and the 2nd, refer to P.5-17 "Gain Switching Function" of Adjustment.

Pr1.15	Title	Mode of position control switching			Related control mode	P	S	T	F
	Range	0 to 10	Unit	—	Default	0			

Set up the triggering condition of gain switching for position control.

Setup value	Switching condition	Gain switching condition
0	Fixed to 1st gain	Fixed to the 1st gain (Pr1.00 to Pr1.04).
1	Fixed to 2nd gain	Fixed to the 2nd gain (Pr1.05 to Pr1.09).
2	With gain switching input	• 1st gain when the gain switching input (GAIN) is open. • 2nd gain when the gain switching input (GAIN) is connected to COM-. * If no input signal is allocated to the gain switching input (GAIN), the 1st gain is fixed.
3	Torque command is large	• Shift to the 2nd gain when the absolute value of the torque command exceeded (level + hysteresis) (%) previously with the 1st gain. • Return to the 1st gain when the absolute value of the torque command was kept below (level - hysteresis) (%) previously during delay time with the 2nd gain.
4	Variation of speed command is large	• Valid only during velocity control. • Shift to the 2nd gain when the absolute value of the speed command variations exceeded (level + hysteresis) (10r/min/s) previously with the 1st gain. • Return to the 1st gain when the absolute value of the speed command variations was kept below (level - hysteresis) (10r/min/s) during delay time previously with the 2nd gain. * The 1st gain is fixed while the velocity control is not applied.
5	Speed command is large	• Valid for position, velocity and full-closed controls. • Shift to the 2nd gain when the absolute value of the speed command exceeded (level + hysteresis) (r/min) previously with the 1st gain. • Return to the 1st gain when the absolute value of the speed command was kept below (level - hysteresis) (r/min) previously during delay time with the 2nd gain.
6	Position deviation is large	• Valid for position and full-closed controls. • Shift to the 2nd gain when the absolute value of the positional deviation exceeded (level + hysteresis) (pulse) previously with the 1st gain. • Return to the 1st gain when the absolute value of the positional deviation was kept below (level - hysteresis) (pulse) previously over delay time with the 2nd gain. * Unit of level and hysteresis (pulse) is set as the encoder resolution for positional control and feedback scale resolution for full-closed control.
7	Position command exists	• Valid for position and full-closed controls. • Shift to the 2nd gain when the positional command was not 0 previously with the 1st gain. • Return to the 1st gain when the positional command was kept 0 previously during delay time with the 2nd gain.
8	Not in positioning complete	• Valid for position and full-closed controls. • Shift to the 2nd gain when the positioning was not completed previously with the 1st gain. • Return to the 1st gain when the positioning was kept in completed condition previously during delay time with the 2nd gain.
9	Actual speed is large	• Valid for position and full-closed controls. • Shift to the 2nd gain when the absolute value of the actual speed exceeded (level + hysteresis) (r/min) previously with the 1st gain. • Return to the 1st gain when the absolute value of the actual speed was kept below (level - hysteresis) (r/min) previously during delay time with the 2nd gain.
10	Position command exists + Actual speed	• Valid for position and full-closed controls. • Shift to the 2nd gain when the positional command was not 0 previously with the 1st gain. • Return to the 1st gain when the positional command was kept at 0 during the delay time and the absolute value of actual speed was kept below (level - hysteresis) (r/min) previously with the 2nd gain.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr1.16	Title	Delay time of position control switching			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	0.1ms	Default	50			

For position controlling : When shifting from the 2nd gain to the 1st gain with Pr1.15 Position control switching mode set at 3, 5, 6, 7, 8, 9 or 10, set up the delay time from trigger detection to the switching operation.

Pr1.17	Title	Level of position control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	50			

For position controlling: Set up triggering level when Pr1.15 Position control gain switching mode is set at 3, 5, 6, 9 or 10.
Unit of setting varies with switching mode.

Caution Set the level equal to or higher than the hysteresis.

Pr1.18	Title	Hysteresis at position control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	33			

For position controlling: Set up triggering hysteresis when Pr1.15 Position control gain switching mode is set at 3, 5, 6, 9 or 10.
Unit of setting varies with switching mode.

Caution When level < hysteresis, the hysteresis is internally adjusted so that it is equal to level.

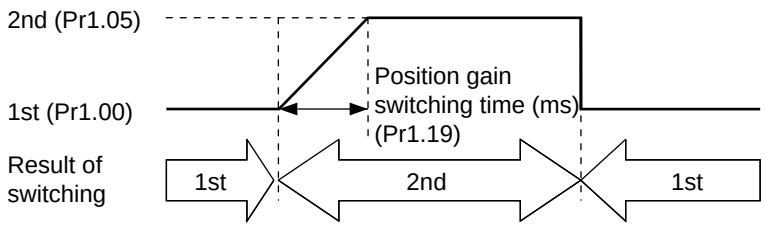
Pr1.19	Title	Position gain switching time			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	Mode-dependent	Default	33			

For position controlling: If the difference between Pr1.00 1st gain of position loop and Pr1.05 2nd gain of poison loop is large, the increasing rate of position loop gain can be limited by this parameter.
The position loop gain will increase over the time set.

<Position gain switching time>
When using position control and full-closed control, gain of position loop rapidly changes, causing torque change and vibration. By adjusting Pr1.19 Position gain switching time, increasing rate of the poison loop gain can be decreased and vibration level can be reduced.

Caution Setting of this parameter does not affect the gain switching time when the gain of position loop is switched to lower level (gain is switched immediately).

Example: 1st (Pr1.00) > 2nd (Pr1.05)



Pr1.20	Title	Mode of velocity control switching			Related control mode	P	S	T	F
	Range	0 to 5	Unit	—	Default	0			

For velocity controlling: Set the condition to trigger gain switching.

Setup value	Switching condition
0	Fixed to the 1st gain.
1	Fixed to the 2nd gain.
2	Gain switching input
3	Torque command
4	Speed command variation is larger.
5	Command speed is larger

Related page For the switching level and timing, refer to P.5-18, "Setup of Gain Switching Condition" of Adjustment.

Pr1.21	Title	Delay time of velocity control switching			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	0.1ms	Default	0			

For velocity controlling: When shifting from the 2nd gain to the 1st gain with Pr1.20 Velocity control switching mode set at 3, 4 or 5, set the delay time from trigger detection to the switching operation.

Pr1.22	Title	Level of velocity control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	0			

For velocity controlling: Set up triggering level when Pr1.20 Velocity control gain switching mode is set at 3, 4 or 5.

Caution Unit of setting varies with switching mode.
Set the level equal to or higher than the hysteresis.

Pr1.23	Title	Hysteresis at velocity control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	0			

For velocity controlling: Set up triggering hysteresis when Pr1.20 Velocity control gain switching mode is set at 3, 4 or 5.

Caution Unit of setting varies with switching mode.
When level < hysteresis, the hysteresis is internally adjusted so that it is equal to level.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr1.24	Title	Mode of torque control switching			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

For torque controlling: Set the condition to trigger gain switching.

Setup value	Switching condition
0	Fixed to the 1st gain.
1	Fixed to the 2nd gain.
2	Gain switching input
3	Torque command

Pr1.25	Title	Delay time of torque control switching			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	0.1ms	Default	0			

For torque controlling : When shifting from the 2nd gain to the 1st gain with Pr1.24 Torque control switching mode set at 3, set up the delay time from trigger detection to the switching operation.

Pr1.26	Title	Level of torque control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	0			

For torque controlling: Set up triggering level when Pr1.24 Torque control gain switching mode is set at 3.
Unit varies depending on the setup of mode of control switching.

Caution Set the level equal to or higher than the hysteresis.

Pr1.27	Title	Hysteresis at torque control switching			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	Mode-dependent	Default	0			

For torque controlling: Set up triggering hysteresis when Pr1.24 Torque control gain switching mode is set at 3.

Caution Unit of setting varies with switching mode.
When level < hysteresis, the hysteresis is internally adjusted so that it is equal to level.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

1. Details of parameter

[Class 2] Damping control

Pr2.00	Title	Adaptive filter mode setup			Related control mode	P	S	T	F
	Range	0 to 4	Unit	—	Default	0			

Set up the resonance frequency to be estimated by the adaptive filter and specify the operation after estimation.

Setup value	Content	
0	Adaptive filter: invalid	Parameters related to the 3rd and 4th notch filter hold the current value.
1	Adaptive filter: 1 filter is valid	One adaptive filter is enabled. Parameters related to the 3rd notch filter will be updated based on adaptive performance.
2	Adaptive filter: 2 filters are valid	Two adaptive filters are enabled. Parameters related to the 3rd and 4th notch filters will be updated based on adaptive performance.
3	Resonance frequency measurement mode	Measure the resonance frequency. Result of measurement can be checked with PANATERM. Parameters related to the 3rd and 4th notch filter hold the current value.
4	Clear result of adaptation	Parameters related to the 3rd and 4th notch filter are disabled and results of adaptive operation are cleared.

Pr2.01	Title	1st notch frequency			Related control mode	P	S	T	F
	Range	50 to 5000	Unit	Hz	Default	5000			

Set the center frequency of the 1st notch filter.

Caution The notch filter function will be invalidated by setting up this parameter to "5000".

Pr2.02	Title	1st notch width selection			Related control mode	P	S	T	F
	Range	0 to 20	Unit	—	Default	2			

Set the width of notch at the center frequency of the 1st notch filter.

Caution Higher the setup, larger the notch width you can obtain. Use with default setup in normal operation.

Pr2.03	Title	1st notch depth selection			Related control mode	P	S	T	F
	Range	0 to 99	Unit	—	Default	0			

Set the depth of notch at the center frequency of the 1st notch filter.

Caution Higher the setup, shallower the notch depth and smaller the phase delay you can obtain.

Pr2.04	Title	2nd notch frequency			Related control mode	P	S	T	F
	Range	50 to 5000	Unit	Hz	Default	5000			

Set the center frequency of the 2nd notch filter.

Caution The notch filter function will be invalidated by setting up this parameter to "5000".

Pr2.05	Title	2nd notch width selection			Related control mode	P	S	T	F
	Range	0 to 20	Unit	—	Default	2			

Set the width of notch at the center frequency of the 2nd notch filter.

Caution

Higher the setup, larger the notch width you can obtain. Use with default setup in normal operation.

Pr2.06	Title	2nd notch depth selection			Related control mode	P	S	T	F
	Range	0 to 99	Unit	—	Default	0			

Set the depth of notch at the center frequency of the 2nd notch filter.

Caution

Higher the setup, shallower the notch depth and smaller the phase delay you can obtain.

Pr2.07	Title	3rd notch frequency			Related control mode	P	S	T	F
	Range	50 to 5000	Unit	Hz	Default	5000			

Notch frequency is automatically set to the 1st resonance frequency estimated by the adaptive filter.

Caution

In no resonance point is found, the frequency is set to 5000.

Pr2.08	Title	3rd notch width selection			Related control mode	P	S	T	F
	Range	0 to 20	Unit	—	Default	2			

Set the width of notch at the center frequency of the 3rd notch filter.

Caution

Higher the setup, larger the notch width you can obtain. Use with default setup in normal operation.
When the applicable filter function is used, parameter value is automatically set.

Pr2.09	Title	3rd notch depth selection			Related control mode	P	S	T	F
	Range	0 to 99	Unit	—	Default	0			

Set the depth of notch at the center frequency of the 3rd notch filter.

Caution

Higher the setup, shallower the notch depth and smaller the phase delay you can obtain.
When the applicable filter function is used, parameter value is automatically set.

Pr2.10	Title	4th notch frequency			Related control mode	P	S	T	F
	Range	50 to 5000	Unit	Hz	Default	5000			

Notch frequency is automatically set to the 2nd resonance frequency estimated by the adaptive filter.

Caution

The notch filter function will be invalidated by setting up this parameter to "5000".

Note

- **For parameters which No. have a suffix of " ∗ "**, changed contents will be validated when you turn on the control power.

Related page

- P.3-30... "Inputs and outputs on connector X4"

Pr2.11	Title	4th notch width selection			Related control mode	P	S	T	F
	Range	0 to 20	Unit	—	Default	2			

Set the width of notch at the center frequency of the 4th notch filter.

Caution

Higher the setup, larger the notch width you can obtain. Use with default setup in normal operation.

When the applicable filter function is used, parameter value is automatically set.

Pr2.12	Title	4th notch depth selection			Related control mode	P	S	T	F
	Range	0 to 99	Unit	—	Default	0			

Set the depth of notch at the center frequency of the 4th notch filter.

Caution

Higher the setup, shallower the notch depth and smaller the phase delay you can obtain.

When the applicable filter function is used, parameter value is automatically set.

Pr2.13	Title	Selection of damping filter switching			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

Among 4 filters select the filters to be used for damping control.

- When setup value is 0: Up to 2 filters can be used simultaneously.
- When setup value is 1 or 2: Select the filter with external input(s) (VS-SEL1 and/or VS-SEL2).

Setup value	VS-SEL2	VS-SEL1	1st damping	2nd damping	3rd damping	4th damping
0	—	—	○	○		
1	—		○		○	
	—	○		○		○
2			○			
		○		○		
	○				○	
	○	○				○

- With setup value 3: Select the filter with command direction.

Setup value	Position command direction	1st damping	2nd damping	3rd damping	4th damping
3	Positive direction	○		○	
	Negative direction		○		○

Pr2.14	Title	1st damping frequency			Related control mode	P	S	T	F
	Range	0 to 2000	Unit	0.1Hz	Default	0			

Pr2.16	Title	2nd damping frequency			Related control mode	P	S	T	F
	Range	0 to 2000	Unit	0.1Hz	Default	0			

Pr2.18	Title	3rd damping frequency			Related control mode	P	S	T	F
	Range	0 to 2000	Unit	0.1Hz	Default	0			

Pr2.20	Title	4th damping frequency			Related control mode	P	S	T	F
	Range	0 to 2000	Unit	0.1Hz	Default	0			

You can set up the 1st to 4th damping frequency of the damping control which suppress vibration at the load edge.
The driver measures vibration at load edge. Setup unit is 0.1[Hz].

Related page The setup frequency is 1.0 to 200.0[Hz]. Setup of 0 to 9 becomes invalid. Refer to P.5-20, "Damping control" as well before using this parameter.

Pr2.15	Title	1st damping filter setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1Hz	Default	0			

Pr2.17	Title	2nd damping filter setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1Hz	Default	0			

Pr2.19	Title	3rd damping filter setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1Hz	Default	0			

Pr2.21	Title	4th damping filter setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1Hz	Default	0			

If torque saturation occurs with damping frequency (1st- 4th) enabled, decrease the setup value, or if the operation is slow, increase it. Usually set it to 0.
The maximum setup value is limited to the applicable damping frequency, and minimum value is limited to a value so that the damping frequency + damping filter setting is equal to or higher than 100.

Caution The maximum setup value is internally limited to the compatible damping frequency..

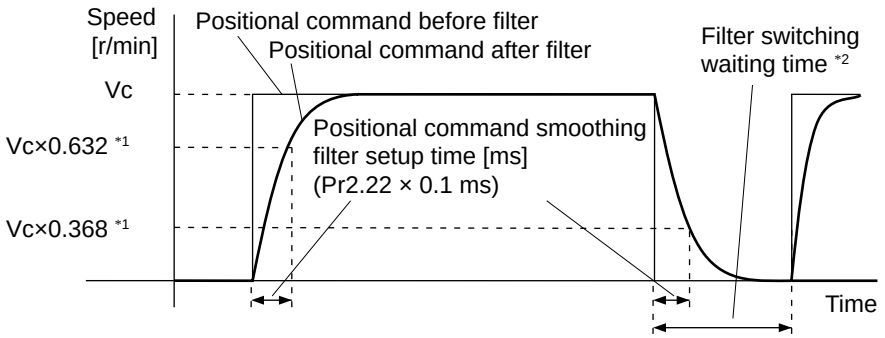
Related page Refer to P.5-24, "Damping control" as well before using this parameter.

Note • **For parameters which No. have a suffix of " ∗ "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr2.22	Title	Positional command smoothing filter			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	0.1ms	Default	0			

- Set up the time constant of the 1st delay filter in response to the positional command.
- When a square wave command for the target speed Vc is applied, set up the time constant of the 1st delay filter as shown in the figure below.



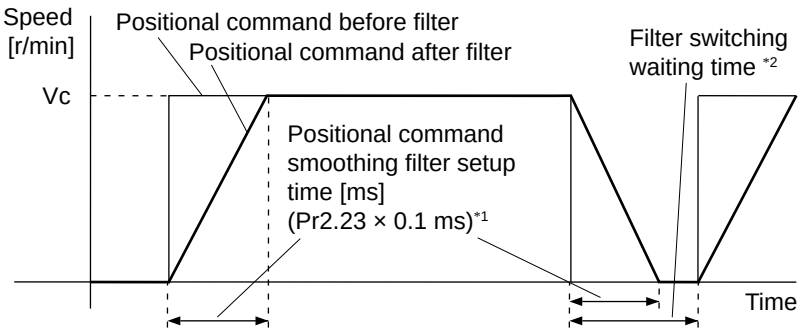
*1 Actual filter time constant (setup value × 0.1 ms) has the maximum absolute error of 0.2 ms for a time constant below 100 ms and the maximum relative error of 0.1% for a time constant 20 ms or more.

*2 Switching of Pr2.22 Positional command smoothing filter is performed on the rising edge of the command with the number of command pulses/0.166 ms is changed from 0 to a value other than 0 while the positioning complete is being output.
If the filter time constant is decreased and positioning complete range is increased, and a many number of plusses are accumulated in the filter (the area equivalent of "value of positional command filter - value of positional command after filter" integrated over the time), at the time of switching, these pulses are discharged at a higher rate, causing the motor to return to the previous position - the motor runs at a speed higher than the command speed for a short time.

*3 Even if Pr2.22 Positional command smoothing filter is changed, it is not applied immediately. If the switching as described in *2 occurs during this delay time, the change of Pr2.22 will be suspended.

Pr2.23	Title	Positional command FIR filter			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	0.1ms	Default	0			

- Set up the time constant of the 1st delay filter in response to the positional command.
- When a square wave command of the target speed Vc is applied, set up the VC arrival time as shown in the figure below.



*1 The actual average travel time (setup value × 0.1 ms) has the maximum absolute error of 0.1 ms for a time constant below 10 ms and the maximum relative error of 1.6% for a time constant 10 ms or more.

*2 When changing Pr2.23 Positional command FIR filter, stop the command pulse and wait until the filter switching wait time has elapsed. The filter switching wait time is the setup value × 0.1 ms + 0.25 ms when the setup time is 10 ms, and setup value × 0.1 ms × 1.05 when the setup time is 10 ms or more. If Pr2.23 is changed while the command pulse is being input, the change is not reflected until the command pulse-less state has continued for the filter switching wait time.

*3 Even if Pr2.23 Positional command FIR filter is changed, it is not applied immediately. If the switching as described in *2 occurs during this delay time, the change of Pr2.23 will be suspended.

Pr3.00	Title	Speed setup, Internal/External switching			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

This driver is equipped with internal speed setup function so that you can control the speed with contact inputs only.

Setup value	Speed setup method
0	Analog speed command (SPR)
1	Internal speed command 1st to 4th speed (Pr3.04 to Pr3.07)
2	Internal speed command 1st to 3rd speed (Pr3.04 to Pr3.06), Analog speed command (SPR)
3	Internal speed command 1st to 8th speed (Pr3.04 to Pr3.11)

<Relationship between Pr3.00 Internal/external switching speed setup and the internal command speed selection 1, 2 and 3, and speed command to be selected>

Setup value	Selection 1 of internal command speed (INTSPD1)	Selection 2 of internal command speed (INTSPD2)	Selection 3 of internal command speed (INTSPD3)	Selection of Speed command
1	OFF	OFF	No effect	1st speed
	ON	OFF		2nd speed
	OFF	ON		3rd speed
	ON	ON		4th speed
2	OFF	OFF	No effect	1st speed
	ON	OFF		2nd speed
	OFF	ON		3rd speed
	ON	ON		Analog speed command
3	The same as Pr3.00=1		OFF	1st to 4th speed
	OFF	OFF	ON	5th speed
	ON	OFF	ON	6th speed
	OFF	ON	ON	7th speed
	ON	ON	ON	8th speed

Pr3.01	Title	Speed command rotational direction selection			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Select the Positive/Negative direction specifying method.

Setup value	Select speed command sign (1st to 8th speed)	Speed command direction (VC-SIGN)	Position command direction
0	+	No effect	Positive direction
	–	No effect	Negative direction
1	Sign has no effect.	OFF	Positive direction
	Sign has no effect.	ON	Negative direction

Note • **For parameters which No. have a suffix of " ∗ "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

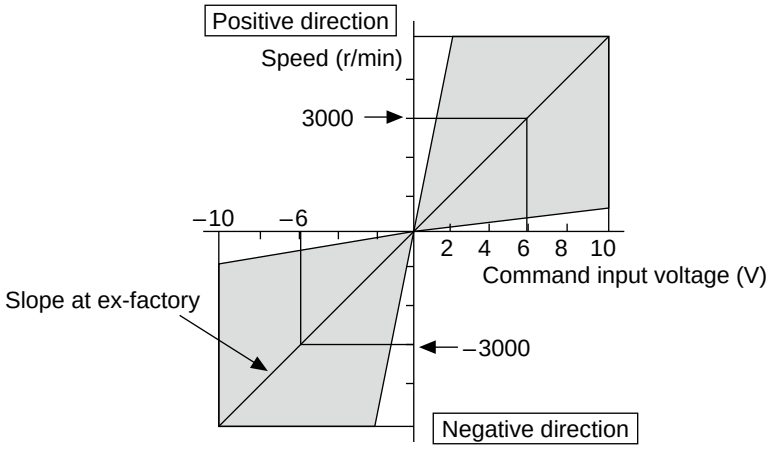
Pr3.02	Title	Input gain of speed command			Related control mode	P	S	T	F
	Range	10 to 2000	Unit	(r/min)/V	Default	500			

Based on the voltage applied to the analog speed command (SPR), set up the conversion gain to motor command speed.

- You can set up a "slope" of the relation between the command input voltage and the motor speed, with Pr3.02.
- Default is set to Pr3.02=500 [r/min], hence input of 6V becomes 3000r/min.

Caution

1. Do not apply more than ±10V to the speed command input (SPR).
2. When you compose a position loop outside of the driver while you use the driver in velocity control mode, the setup of Pr3.02 gives larger variance to the overall servo system.
   Pay an extra attention to oscillation caused by larger setup of Pr3.02.



Pr3.03	Title	Reversal of speed command input			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	1			

Specify the polarity of the voltage applied to the analog speed command (SPR).

Setup value	Motor rotating direction	
0	Non-reversal	"+Voltage" → "Positive direction", "–Voltage" → "Negative direction"
1	Reversal	"+Voltage" → "Negative direction", "–Voltage" → "Positive direction"

Note Default of this parameter is 1, and the motor turns to CW with (+) signal, this has compatibility to existing MINAS series driver.

Caution When you compose the servo drive system with this driver set to velocity control mode and external positioning unit, the motor might perform an abnormal action if the polarity of the speed command signal from the unit and the polarity of this parameter setup does not match.

Pr3.04	Title	**1st speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.05	Title	**2nd speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.06	Title	**3rd speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.07	Title	**4th speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.08	Title	**5th speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.09	Title	**6th speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.10	Title	**7th speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			
Pr3.11	Title	**8th speed of speed setup**			Related control mode	P	**S**	T	F
	Range	**–20000 to 20000**	Unit	**r/min**	Default	**0**			

Set up internal command speeds, 1st to 8th.

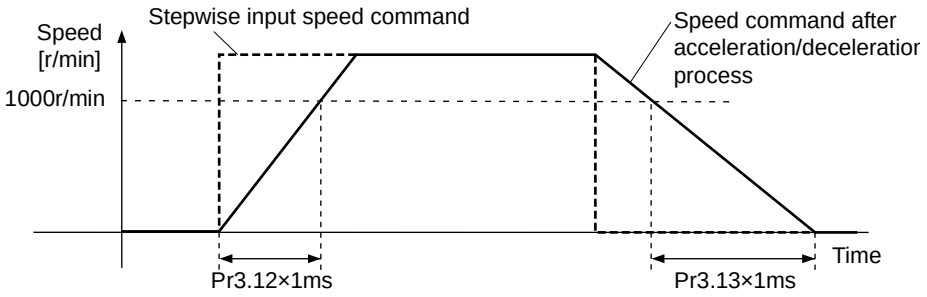
Pr3.12	Title	**Acceleration time setup**			Related control mode	P	**S**	T	F
	Range	**0 to 10000**	Unit	**ms/ (1000r/min)**	Default	**0**			
Pr3.13	Title	**Deceleration time setup**			Related control mode	P	**S**	T	F
	Range	**0 to 10000**	Unit	**ms/ (1000r/min)**	Default	**0**			

Set up acceleration/deceleration processing time in response to the speed command input. Set the time required for the speed command (stepwise input) to reach 1000 r/min to Pr3.12 Acceleration time setup. Also set the time required for the speed command to reach from 1000 r/min to 0 r/min, to Pr3.13 Deceleration time setup.

Assuming that the target value of the speed command is Vc(r/min), the time required for acceleration/deceleration can be computed from the formula shown below.

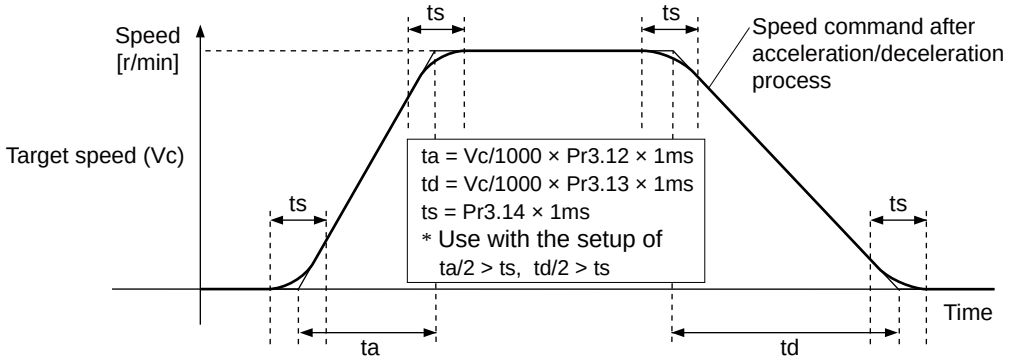
Acceleration time (ms) = Vc/1000 × Pr3.12 × 1 ms

Deceleration time (ms) = Vc/1000 × Pr3.13 × 1 ms



Pr3.14	Title	Sigmoid acceleration/ deceleration time setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	ms	Default	0			

Set S-curve time for acceleration/deceleration process when the speed command is applied. According to Pr3.12 Acceleration time setup and Pr3.13 Deceleration time setup, set up sigmoid time with time width centering the inflection point of acceleration/deceleration.



Pr3.15	Title	Speed zero-clamp function selection			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

You can set up the function of the speed zero clamp input.

Setup value	Function of ZEROSPD (Pin-26)
0	Invalid: Speed zero-clamp input is ignored.
1	Speed command is forced to 0 when the speed zero clamp (ZEROSPD) input signal is turned ON.
2	Speed command is forced to 0 when the speed zero clamp (ZEROSPD) input signal is turned ON. And when the actual motor speed drops to Pr3.16 Speed zero clamp level or below, the position control is selected and servo lock is activated at this point. The fundamental operations except for this function (switching to the position control) are identical to those when setup value is 1.
3	When the speed zero clamp (ZEROSPD) input signal is ON and speed command is below Pr3.16 Speed zero clamp level - 10 r/min, then the position control is selected and servo lock is activated at that point.

Pr3.16	Title	Speed zero clamp level			Related control mode	P	S	T	F
	Range	10 to 20000	Unit	r/min	Default	30			

Select the timing at which the position control is activated as the Pr3.15 Speed zero-clamp function selection is set to 2 or 3.
If Pr3.15 = 3, then hysteresis of 10 r/min is provided for detection.

Note
- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page
- P.3-30... "Inputs and outputs on connector X4"

Pr3.17

Title	Selection of torque command			Related control mode	P S **T** F
Range	**0 to 2**	Unit	—	Default	**0**

You can select the input of the torque command and the speed limit.

Setup value	Torque command input	Velocity limit input
0	Analog input 1 *1 (AI1, 16-bit resolution)	Parameter value (Pr3.25)
1	Analog input 2 (AI2, 12-bit resolution)	Analog input 1 (AI1, 16-bit resolution)
2	Analog input 1 *1 (AI1, 16-bit resolution)	Parameter value (Pr3.25, Pr3.26)

*1 For Pr0.01 Control mode setup = 5 (velocity/torque control), the torque command input is the analog input 2 (AI2, 12-bit resolution).

Pr3.18

Title	Torque command direction selection			Related control mode	P S **T** F
Range	**0 to 1**	Unit	—	Default	**0**

Select the direction positive/negative direction of torque command.

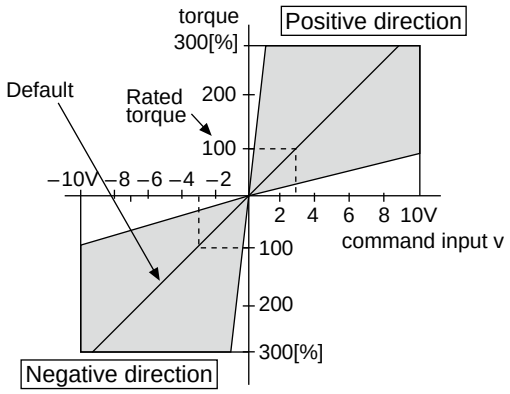
Setup value	Designation
0	Specify the direction with the sign of torque command. Example: Torque command input (+) for positive direction, (–) for negative direction
1	Specify the direction with torque command sign (TC-SIGN). OFF: Positive direction, ON: Negative direction

Pr3.19

Title	Input gain of torque command			Related control mode	P S **T** F
Range	**10 to 100**	Unit	**0.1V/100%**	Default	**30**

Based on the voltage (V) applied to the analog torque command (TRQ R), set up the conversion gain to torque command (%).

- Unit of the setup value is [0.1V/100%] and set up input voltage necessary to produce the rated torque.
- Default setup of 30 represents 3V/100%.



Pr3.20

Title	Input reversal of torque command			Related control mode	P S **T** F
Range	**0 to 1**	Unit	—	Default	**0**

Set up the polarity of the voltage applied to the analog torque command (TRQR).

Setup value	Direction of motor output torque	
0	Non-reversal	“+Voltage” → “Positive direction”, “–Voltage” → “Negative direction”
1	Reversal	“+Voltage” → “Negative direction”, “–Voltage” → “Positive direction”

Pr3.21	Title	Speed limit value 1			Related control mode	P S **T** F
	Range	**0 to 20000**	Unit	**r/min**	Default	**0**

Set up the speed limit used for torque controlling.
During the torque controlling, the speed set by the speed limit value cannot be exceeded.
When Pr3.17 = 2, the speed limit is applied upon receiving positive direction command.

Pr3.22	Title	Speed limit value 2			Related control mode	P S **T** F
	Range	**0 to 20000**	Unit	**r/min**	Default	**0**

Speed limit value of negative direction command when Pr3.17 = 2.

Pr3.17	Pr3.21	Pr3.22	Pr3.15	Speed zero clamp (ZEROSPD)	Analog torque command direction	Speed limit value
0	0 to 20000	No effect	0	No effect	No effect	Pr3.21 setup value
			1 to 3	OFF		Pr3.21 setup value
				ON		0
2	**0 to 20000**	**0 to 20000**	**0**	No effect	Positive direction	Pr3.21 setup value
					Negative direction	Pr3.22 setup value
	0 to 20000	0 to 20000	1 to 3	OFF	Positive direction	Pr3.21 setup value
					Negative direction	Pr3.22 setup value
	0 to 20000	0 to 20000	1 to 3	ON	No effect	0

Pr3.23 *	Title	External scale selection			Related control mode	P S T **F**
	Range	**0 to 2**	Unit	—	Default	**0**

Select the type of feedback scale.

Setup value	Feedback scale type	Compatible scale	Compatible speed
0	A,B phase output type *1	Feedback scale of A, B phase output type	to 4Mpps (after quadrupled)
1	Serial communication type (incremental version) *1	Sony Manufacturing Systems Corp. SR75, SR85, SL700, SL710	to 400Mpps
2	Serial communication type (absolute version) *1	Mitsutoyo Corp. AT573, ST771A, ST773A Sony Manufacturing Systems Corp. SR77, SR87	to 400Mpps

*1 Connect the feedback scale so that it increments the count as the motor shaft turns positive direction, and decrements as the shaft turns negative direction. If this connection arrangement is impossible due to installation condition, etc., use the count reverse function of Pr3.26 Reversal of direction of external scale.

Caution When the setup value is 1 or 2 while the A, B phase output type is connected, Err50.0 External scale wiring error protection occurs, and if the setup value is 0 while the serial communication type is connected, Err55.0, 1 or 2 A phase, B phase or Z phase wiring error protection will occur.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4" • P.6-3 "Protective Function"

Pr3.24 *	Title	Numerator of external scale division			Related control mode	P	S	T	F
	Range	0 to 2^{20}	Unit	—	Default	0			

Set up the numerator of the feedback scale dividing setup.
When setup value = 0, encoder resolution is used as numerator of the division.

Pr3.25 *	Title	Denominator of external scale division			Related control mode	P	S	T	F
	Range	1 to 2^{20}	Unit	—	Default	10000			

- Check the number of encoder pluses per one motor revolution and the number of feedback scale pulses per one motor revolution, and then set up the numerator of external scale division (Pr3.24) and the denominator of external scale division (Pr3.25) to establish the expression shown below.
- With Pr3.24 set at 0, the encoder resolution is automatically used as numerator.

Example: When ball screw pitch is 10 mm, scale 0.1 _m/pulse, encoder resolution 20 bits (1048.576 pulses);

$$\frac{\text{Pr3.24}\ \boxed{1048576}}{\text{Pr3.25}\ \boxed{100000}} = \frac{\text{Encoder resolution per one motor revolution [pulse]}}{\text{Feedback scale resolution per one motor revolution [pulse]}}$$

Caution If this ratio is wrong, the difference between the position calculated based on the encoder pulses and the position calculated based on the feedback scale pulses becomes large over a long travel distance and will activate the excess hybrid deviation error protection.

Pr3.26 *	Title	Reversal of direction of external scale			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Reverse the direction of feedback scale, feedback counter.

Setup value	Content
0	Count value of external scale can be used as it is.
1	Sign (positive/negative) of count value of external scale should be inverted.

Note For setting method of this parameter, refer to P.3-12 Full closed control mode.

Pr3.27 *	Title	External scale Z phase disconnection detection disable			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Enable/disable Z-phase disconnection detection when A, B phase output type feedback scale is used.

Setup value	Content
0	Valid
1	Invalid

Pr3.28 *	Title	Hybrid deviation excess setup			Related control mode	P	S	T	F
	Range	1 to 2^{27}	Unit	Command unit	Default	16000			

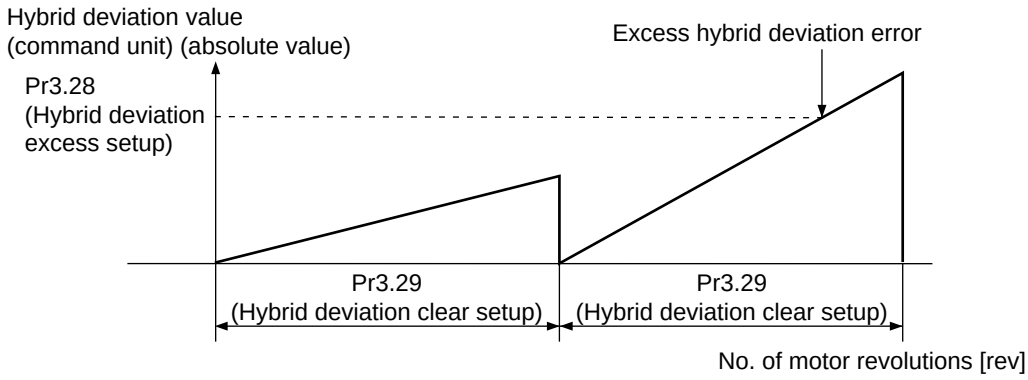
You can setup the permissible gap (hybrid deviation) between the present motor position and the present feedback scale position.

Pr3.29 *	Title	Hybrid deviation clear setup			Related control mode	P	S	T	F
	Range	0 to 100	Unit	Revolution	Default	0			

As the motor turns the number of revolutions set by this parameter, the hybrid deviation is cleared to 0. No clearing is made with setup value 0.

<Hybrid deviation clear feature>

As the motor reaches the number of revolutions set by Pr3.29 Hybrid deviation clear setup, clear the hybrid deviation to 0. This feature allows the motor to be used in an application where hybrid deviation accumulate due to slippage, etc.



Note: Revolution in the hybrid deviation clear setup is counted by using encoder feedback pulses.

Caution To use the hybrid deviation clear, be sure to set Pr3.29 Hybrid deviation clear setup to the appropriate value. If the setup value is too small with respect to the value of Pr3.28 Hybrid deviation excess setup, abnormal operation due to e.g. feedback scale connection error cannot be protected.
Limit sensor should be used to assure safety.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

1. Details of parameter

[Class 4] I/F monitor setting

Pr4.00 *	Title	SI1 input selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00828282h (8553090)			

Assign functions to SI1 inputs.
These parameters are presented in hexadecimals. *5
Hexadecimal presentation is followed by a specific control mode designation.
0 0 – – – * * h : position/full-closed control
0 0 – – * * – – h : velocity control
0 0 * * – – – – h : torque control
Replace * * with the function number.
For the function number see the table below. Logical setup is also a function number.

Title	Symbol	Setup value	
		a-contact	b-contact
Invalid	–	00h	Do not setup.
Positive direction over-travel inhibition input	POT	01h	81h
Negative direction over-travel inhibition input	NOT	02h	82h
Servo-ON input *1	SRV-ON	03h	83h
Alarm clear input	A-CLR	04h	Do not setup.
Control mode switching input *2	C-MODE	05h	85h
Gain switching input	GAIN	06h	86h
Deviation counter clear input *3	CL	07h	Do not setup.
Command pulse inhibition input *4	INH	08h	88h
Torque limit switching input	TL-SEL	09h	89h
Damping control switching input 1	VS-SEL1	0Ah	8Ah
Damping control switching input 2	VS-SEL2	0Bh	8Bh
Electronic gear switching input 1	DIV1	0Ch	8Ch
Electronic gear switching input 2	DIV2	0Dh	8Dh
Selection 1 input of internal command speed	INTSPD1	0Eh	8Eh
Selection 2 input of internal command speed	INTSPD2	0Fh	8Fh
Selection 3 input of internal command speed	INTSPD3	10h	90h
Speed zero clamp input	ZEROSPD	11h	91h
Speed command sign input	VC-SIGN	12h	92h
Torque command sign input	TC-SIGN	13h	93h
Forced alarm input	E-STOP	14h	94h
Inertia ratio switching input	J-SEL	15h	95h

Caution

- Do not setup to a value other than that specified in the table.
- Do not assign specific function to 2 or more signals. Duplicated assignment will cause Err33.0 I/F input multiple assignment error 1 or Err33.1 I/F input multiple assignment error 2.

*1 Servo-on input signal (SRV-ON) must be used to enable servo-on.

*2 When using control mode switching input (C-MODE), set the signal to all control modes. If the signal is set to only 1 or 2 control modes, Err33.2 I/F input function number error 1 or Err33.3 I/F input function number error 2 will be generated.
- The control input pin set to invalid state does not affect any operation.
- Function (servo-on input, alarm clear, etc.) to be used in multiple control modes must be assigned to the same pin with correct logical arrangement. Incorrect setting will cause Err33.0 I/F input multiple assignment error 1 or Err33.1 I/F input multiple assignment error 2.

*3 Deviation counter clear input (CL) can be assigned only to SI7 input. Wrong assignment will cause Err33.6 Counter clear assignment error.

*4 Command pulse inhibit input (INH) can be assigned only to SI10 input. Wrong assignment will cause Err33.7 Command pulse input inhibit input.

*5 Note that the front panel indicates parameter value in decimal number.

Pr4.01 *	Title	**SI2 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00818181h (8487297)**			
Pr4.02 *	Title	**SI3 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**0091910Ah (9539850)**			
Pr4.03 *	Title	**SI4 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00060606h (394758)**			
Pr4.04 *	Title	**SI5 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**0000100Ch (4108)**			
Pr4.05 *	Title	**SI6 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00030303h (197379)**			
Pr4.06 *	Title	**SI7 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00000f07h (3847)**			

Caution Deviation counter clear (CL) can be set up only with this parameter. If any other parameter is used for this purpose, error code No.33.6 Counter clear assignment error will be issued.

Pr4.07 *	Title	**SI8 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00040404h (263172)**			
Pr4.08 *	Title	**SI9 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00050505h (328965)**			
Pr4.09 *	Title	**SI10 input selection**			Related control mode	P	S	T	F
	Range	**0 to 00FFFFFFh**	Unit	—	Default	**00000E88h (3720)**			

Assign functions to SI2 to SI10 inputs.
These parameters are presented in hexadecimals.
Setup procedure is the same as described for Pr4.00.

Caution Command pulse inhibition input (INH) can be setup only with this parameter. If any other parameter is used for this purpose, error code No.33.7 INH assignment error will be issued.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr4.10 *	Title	SO1 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00030303h (197379)			

Assign functions to SO1 outputs.
These parameters are presented in hexadecimals. *1
Hexadecimal presentation is followed by a specific control mode designation.
0 0 – – – – * * h : position/full-closed control
0 0 – – * * – – h : velocity control
0 0 * * – – – – h : torque control
Replace * * with the function number.
For the function number see the table below. Logical setup is also a function number.

Setup value	Title	Symbol
00h	Invalid	–
02h	Servo-Ready output	S-RDY
03h	External brake release signal	BRK-OFF
04h	Positioning complete output	INP
05h	At-speed output	AT-SPPED
06h	Torque in-limit signal output	TLC
07h	Zero-speed detection output signal	ZSP
08h	Speed coincidence output	V-COIN
09h	Alarm output 1	WARN1
0Ah	Alarm output 2	WARN2
0Bh	Positional command ON/OFF output	P-CMD
0Ch	Positioning complete 2	INP2
0Dh	Speed in-limit output	V-LIMIT
0Eh	Alarm attribute output	ALM-ATB
0Fh	Speed command ON/OFF output	V-CMD

• Same function can be assigned to 2 or more output signals.
• Control output pin set to invalid always has the output transistor turned OFF.
• Do not change the setup value shown in the table.

Caution *1 Note that the setup values are displayed in decimal on the front panel.

Pr4.11 *	Title	SO2 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00020202h (131586)			

Pr4.12 *	Title	SO3 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00010101h (65793)			

Pr4.13 *	Title	SO4 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00050504h (328964)			

Pr4.14 *	Title	SO5 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00070707h (460551)			

Pr4.15 *	Title	SO6 output selection			Related control mode	P	S	T	F
	Range	0 to 00FFFFFFh	Unit	—	Default	00060606h (394758)			

Assign functions to SO2 to SO6 outputs.
These parameters are presented in hexadecimals.
Setup procedure is the same as described for Pr4.10.

Pr4.16	Title	Type of analog monitor 1			Related control mode	P	S	T	F
	Range	0 to 21	Unit	—	Default	0			

Select the type of monitor for analog monitor 1. *See the table shown on the next page.

Pr4.18	Title	Type of analog monitor 2			Related control mode	P	S	T	F
	Range	0 to 21	Unit	—	Default	4			

Select the type of monitor for analog monitor 2. *See the table shown on the next page.

Pr4.17	Title	Analog monitor 1 output gain			Related control mode	P	S	T	F
	Range	0 to 214748364	Unit	[Monitor unit in Pr4.16] / V	Default	0			

Set up the output gain of analog monitor 1.
For Pr4.16 = 0 Motor speed, 1 V is output at the motor speed [r/min] = Pr4.17 setup value.

Pr4.19	Title	Analog monitor 2 output gain			Related control mode	P	S	T	F
	Range	0 to 214748364	Unit	[Monitor unit in Pr4.16] / V	Default	0			

Set up the output gain of analog monitor 2.
For Pr4.18 = 3 Torque command, 1 V is output at the torque command [%] = Pr4.18 setup value.

Pr4.20	Title	Type of digital monitor			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

Select type of the digital monitor.

Setup value	Type of monitor	Digital signal output	
		L output	H output
0	Positioning complete condition	Not completed	Completed
1	Positional command	Without command	With command
2	Alarm	Not generated	Generated
3	Gain selected	1st gain	2nd gain (including 3rd gain)

Note
- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

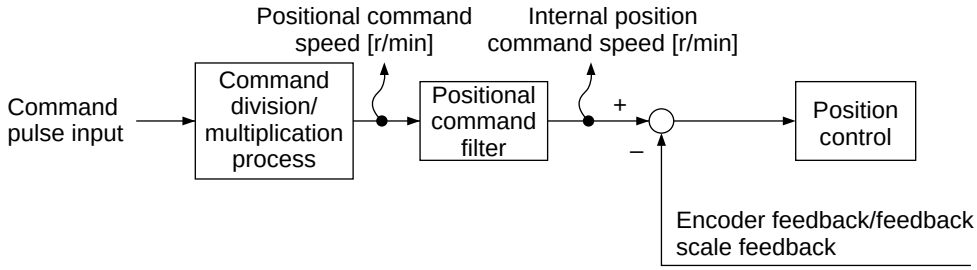
Related page
- P.3-30... "Inputs and outputs on connector X4"

Pr4.16/Pr4.18	Type of monitor	Unit	Output gain for setting Pr4.17/Pr4.19 = 0
0	Motor speed	r/min	500
1	Positional command speed *3	r/min	500
2	Internal positional command speed *3	r/min	500
3	Velocity control command	r/min	500
4	Torque command	%	33
5	Command positional deviation *4	pulse (Command unit)	3000
6	Encoder positional deviation *4	pulse (Encoder unit)	3000
7	Full-closed deviation *4	pulse (External scale unit)	3000
8	Hybrid deviation	pulse (Command unit)	3000
9	Voltage across PN	V	80
10	Regenerative load factor	%	33
11	Overload factor	%	33
12	Positive direction torque limit	%	33
13	Negative direction torque limit	%	33
14	Speed limit value	r/min	500
15	Inertia ratio	%	500
16	Analog input 1 *2	V	1
17	Analog input 2 *2	V	1
18	Analog input 3 *2	V	1
19	Encoder temperature *5	°C	10
20	Driver temperature	°C	10
21	Encoder single-turn data *1	pulse (Encoder unit)	110000

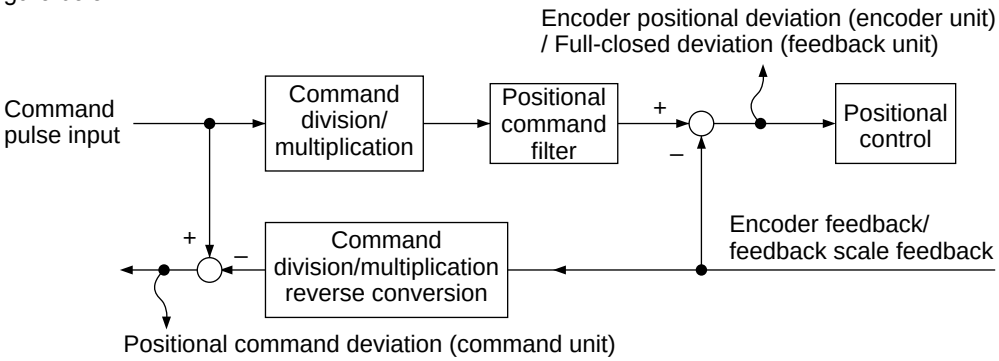
*1 The encoder rotation data CCW is always positive value regardless of Pr0.00 Rotational direction setup. The direction of other monitor data basically follows Pr0.00 Rotational direction setup.

*2 Analog inputs 1, 2 and 3 always output terminal voltage regardless of usage of analog input function.

*3 For the command pulse input, the speed before the command filter (smoothing, FIR filter) is defined as positional command speed and speed after filter is defined as internal command speed.



*4 Command positional deviation is the deviation with respect to the command pulse input and the encoder positional deviation/ full-closed positional deviation is the deviation at the input section of the positional control, as described in the figure below.



*5 Temperature information from the encoder includes value only when it is a 20-bit incremental encoder. Otherwise, the value is always 0.

Pr4.21	Title	Analog monitor output setup			Related control mode	P	S	T	F
	Range	0 to 2	Unit	—	Default	0			

Select output format of the analog monitor.

Setup value	Output format	
0	Signed data output	–10 V to 10 V
1	Absolute value data output	0 V to 10 V
2	Data output with offset	0 V to 10 V (5 V at center)

Pr4.22	Title	Analog input 1 (AI1) offset setup			Related control mode	P	S	T	F
	Range	–5578 to 5578	Unit	0.359mV	Default	0			

Set up the offset correction value applied to the voltage fed to the analog input 1.

Pr4.23	Title	Analog input 1 (AI1) filter			Related control mode	P	S	T	F
	Range	0 to 6400	Unit	0.01ms	Default	0			

Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 1.

Pr4.24	Title	Analog input 1 (AI1) overvoltage setup			Related control mode	P	S	T	F
	Range	0 to 100	Unit	0.1V	Default	0			

Set up the excessive level of the input voltage of analog input 1 by using the voltage associated with offset.

Pr4.25	Title	Analog input 2 (AI2) offset setup			Related control mode	P	S	T	F
	Range	–342 to 342	Unit	5.86mV	Default	0			

Set up the offset correction value applied to the voltage fed to the analog input 2.

Pr4.26	Title	Analog input 2 (AI2) filter			Related control mode	P	S	T	F
	Range	0 to 6400	Unit	0.01ms	Default	0			

Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 2.

Pr4.27	Title	Analog input 2 (AI2) overvoltage setup			Related control mode	P	S	T	F
	Range	0 to 100	Unit	0.1V	Default	0			

Set up the excessive level of the input voltage of analog input 2 by using the voltage associated with offset.

Note

- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page
- P.3-30... "Inputs and outputs on connector X4"

Pr4.28	Title	Analog input 3 (AI3) offset setup			Related control mode	P	S	T	F
	Range	–342 to 342	Unit	5.86mV	Default	0			

Set up the offset correction value applied to the voltage fed to the analog input 3.

Pr4.29	Title	Analog input 3 (AI3) filter			Related control mode	P	S	T	F
	Range	0 to 6400	Unit	0.01ms	Default	0			

Set up the time constant of 1st delay filter that determines the lag time behind the voltage applied to the analog input 3.

Pr4.30	Title	Analog input 3 (AI3) overvoltage setup			Related control mode	P	S	T	F
	Range	0 to 100	Unit	0.1V	Default	0			

Set up the excessive level of the input voltage of analog input 3 by using the voltage associated with offset.

Pr4.31	Title	Positioning complete (In-position) range			Related control mode	P	S	T	F
	Range	0 to 262144	Unit	Command unit	Default	10			

Set up the timing of positional deviation at which the positioning complete signal (INP1) is output.

Caution

The command unit is used as the default unit but can be replaced by the encoder unit by using Pr5.20. Positioning unit selection. Note that when the encoder unit is used, unit of Pr0.14 Positional deviation excess setup is also changed.

Pr4.32	Title	Positioning complete (In-position) output setup			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

Select the condition to output the positioning complete signal (INP1).

Setup value	Action of positioning complete signal
0	The signal will turn on when the positional deviation is smaller than Pr4.31 (Positioning complete range)
1	The signal will turn on when there is no position command and the positional deviation is smaller than Pr4.31 (Positioning complete range).
2	The signal will turn on when there is no position command, the zero-speed detection signal is ON and the positional deviation is smaller than Pr4.31 (Positioning complete range).
3	The signal will turn on when there is no position command and the positional deviation is smaller than Pr4.31 (Positioning complete range). Then holds "ON" status until the next position command is entered.Subsequently, ON state is maintained until Pr4.33 INP hold time has elapsed. After the hold time, INP output will be turned ON/OFF according to the coming positional command or condition of the positional deviation.

Pr4.33	Title	INP hold time			Related control mode	P	S	T	F
	Range	0 to 30000	Unit	1ms	Default	0			

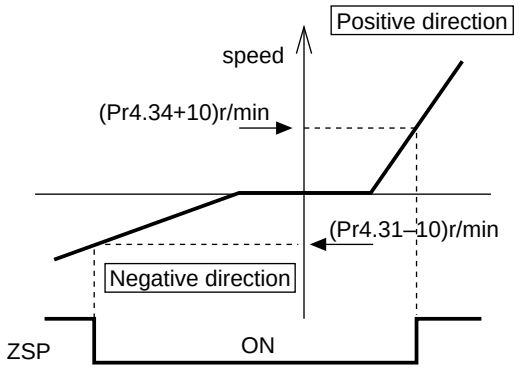
Set up the hold time when Pr4.32 Positioning complete output setup = 13.

Setup value	State of positioning complete signal
0	The hold time is maintained definitely, keeping ON state until the next positional command is received.
1 to 30000	ON state is maintained for setup time (ms) but switched to OFF state as the positional command is received during hold time.

Pr4.34	Title	Zero-speed			Related control mode	P	S	T	F
	Range	10 to 20000	Unit	r/min	Default	50			

You can set up the timing to feed out the zero-speed detection output signal (ZSP or TCL) in rotational speed [r/min].
The zero-speed detection signal (ZSP) will be fed out when the motor speed falls below the setup of this parameter, Pr4.34.
Output the speed coincidence (V-COIN) when the difference between the speed command and the motor speed reduces to a value equal to or below the speed specified by this parameter Pr4.34.

- The setup of Pr61 is valid for both Positive and Negative direction regardless of the motor rotating direction.
- There is hysteresis of 10 [r/min].

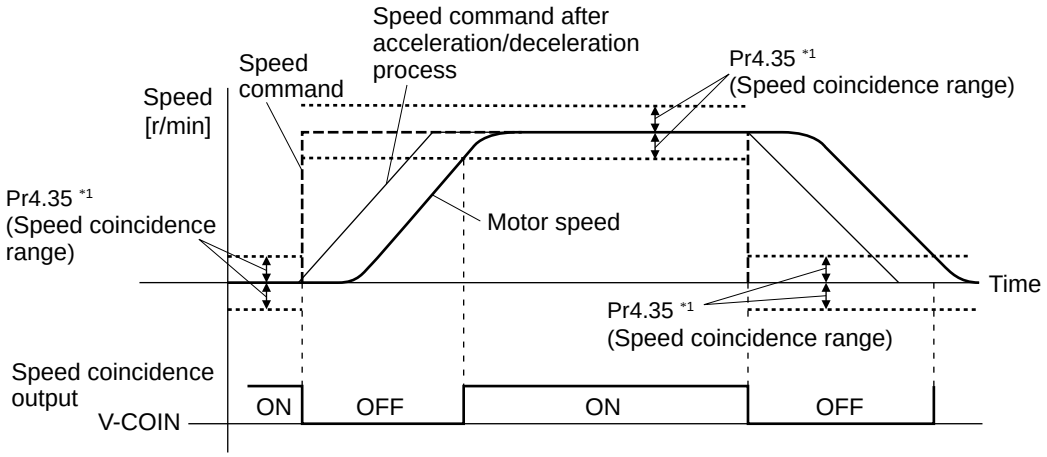


Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr4.35	Title	Speed coincidence range			Related control mode	P	S	T	F
	Range	10 to 20000	Unit	r/min	Default	50			

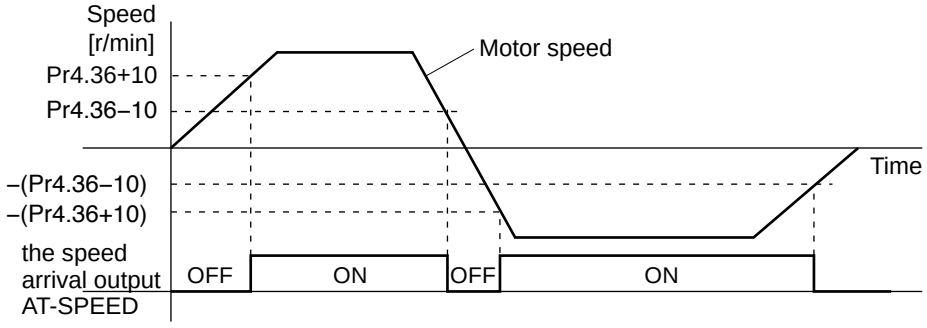
Set the speed coincidence (V-COIN) output detection timing.
Output the speed coincidence (V-COIN) when the difference between the speed command and the motor speed is equal to or smaller than the speed specified by this parameter.



*1 Because the speed coincidence detection is associated with 10 r/min hysteresis, actual detection range is as shown below.
Speed coincidence output OFF → ON timing (Pr4.35 – 10) r/min
Speed coincidence output ON → OFF timing (Pr4.35 + 10) r/min

Pr4.36	Title	At-speed (Speed arrival)			Related control mode	P	S	T	F
	Range	10 to 20000	Unit	r/min	Default	1000			

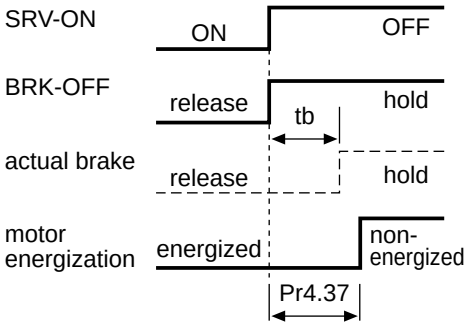
Set the detection timing of the speed arrival output (AT-SPEED).
When the motor speed exceeds this setup value, the speed arrival output (AT-SPEED) is output.
Detection is associated with 10 r/min hysteresis.



Pr4.37	Title	Mechanical brake action at stalling setup			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	1ms	Default	0			

You can set up the time from when the brake release signal (BRK-OFF) turns off to when the motor is de-energized (Servo-free), when the motor turns to Servo-OFF while the motor is at stall.

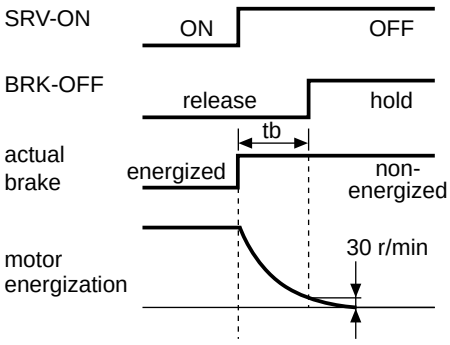
- Set up to prevent a micro-travel/ drop of the motor (work) due to the action delay time (tb) of the brake
- After setting up Pr6a >= tb, then compose the sequence so as the driver turns to Servo-OFF after the brake is actually activated.



Pr4.38	Title	Mechanical brake action at running setup			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	1ms	Default	0			

You can set up time from when detecting the off of Servo-ON input signal (SRV-ON) is to when external brake release signal (BRK-OFF) turns off, while the motor turns to servo off during the motor in motion.

- Set up to prevent the brake deterioration due to the motor running.
- At Servo-OFF during the motor is running, tb of the right fig. will be a shorter one of either Pr6B setup time, or time lapse till the motor speed falls below 30r/min.



Pr4.39	Title	Brake release speed setup			Related control mode	P	S	T	F
	Range	30 to 3000	Unit	r/min	Default	30			

Set up the speed timing of brake output checking during operation.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr4.40	Title	Selection of alarm output 1			Related control mode	P	S	T	F
	Range	0 to 10	Unit	—	Default	0			

Pr4.41	Title	Selection of alarm output 2			Related control mode	P	S	T	F
	Range	0 to 10	Unit	—	Default	0			

Select the type of alarm issued as the alarm output 1 or 2.

Setup value	Alarm	Content
0	—	ORed output of all alarms.
1	Overload protection	Load factor is 85% or more the protection level.
2	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.
3	Battery alarm	Battery voltage is 3.2 V or lower.
4	Fan alarm	Fan has stopped for 1 sec.
5	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.
6	Encoder overheat alarm	The encoder detects overheat alarm.
7	Oscillation detection alarm	Oscillation or vibration is detected.
8	Lifetime detection alarm	Life expectancy of capacitor or fan becomes short.
9	External scale error alarm	The feedback scale detects the alarm.
10	External scale communication alarm	The number of successive feedback scale communication errors exceeds the specified value.

Related page For detailed description of alarm types, refer to P.3-46.

Pr4.42	Title	2nd Positioning complete (In-position) range			Related control mode	P	S	T	F
	Range	0 to 262144	Unit	Command unit	Default	10			

The INP2 turns ON whenever the positional deviation is lower than the value set up in this parameter, without being affected by Pr4.32 Positioning complete output setup. (Presence/absence of positional command is not related to this judgment.)

Caution The command unit is used as the default unit but can be replaced by the encoder unit by using Pr5.20. Positioning unit selection. Note that when the encoder unit is used, unit of Pr0.14 Positional deviation excess setup is also changed.

1. Details of parameter

[Class 5] Enhancing setting

Pr5.00	Title	2nd numerator of electronic gear			Related control mode	P S T F
	Range	0 to 2^{30}	Unit	—	Default	0

Pr5.01	Title	3rd numerator of electronic gear			Related control mode	P S T F
	Range	0 to 2^{30}	Unit	—	Default	0

Pr5.02	Title	4th numerator of electronic gear			Related control mode	P S T F
	Range	0 to 2^{30}	Unit	—	Default	0

Set the 2nd to 4th numerator of division/multiplication operation made according to the command pulse input.
This setup is enabled when Pr0.08 command pulse counts per one motor revolution = 0.
When the setup value is 0, the numerator is replaced by the encoder resolution.

Pr5.03 *	Title	Denominator of pulse output division			Related control mode	P S T F
	Range	10 to 262144	Unit	—	Default	0

For details, refer to P.4-9 Pr0.11.

Pr5.04 *	Title	Over-travel inhibit input setup			Related control mode	P S T F
	Range	0 to 2	Unit	—	Default	1

Set up the operation of the run-inhibition (POT, NOT) inputs.

Setup value	Operation
0	POT → Inhibit positive direction travel NOT → Inhibit negative direction travel
1	Disable POT, NOT
2	POT or NOT input activates Err38.0 Run-inhibition input protection.

Pr5.05 *	Title	Sequence at over-travel inhibit			Related control mode	P S T F
	Range	0 to 2	Unit	—	Default	0

When Pr5.04 Over-travel inhibition = 0, specify the status during deceleration and stop after application of the over-travel inhibition (POT, NOT).

<Details of Pr5.05 (Sequence at over-travel inhibit)>

Pr5.04	Pr5.05	During deceleration	After stalling	Deviation counter content
0	**0**	Dynamic brake action	Torque command=0 towards inhibited direction	Hold
	1	Torque command=0 towards inhibited direction	Torque command=0 towards inhibited direction	Hold
	2	Emergency stop	Torque command=0 towards inhibited direction	Clears before/ after deceleration

Note

- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page
- P.3-30... "Inputs and outputs on connector X4"
- P.6-3 "Protective Function"

Pr5.06	Title	Sequence at Servo-Off			Related control mode	P	S	T	F
	Range	0 to 9	Unit	—	Default	0			

Specify the status during deceleration and after stop, after servo-off.

Setup value	During deceleration *3	After stalling	Positional deviation/ feedback scale deviation
0	Dynamic Brake (DB) action	Dynamic Brake (DB) action	Clear *4
1	Free-run (DB OFF)	Dynamic Brake (DB) action	Clear *4
2	Dynamic Brake (DB) action	Free-run (DB OFF)	Clear *4
3	Free-run (DB OFF)	Free-run (DB OFF)	Clear *4
4	Dynamic Brake (DB) action	Dynamic Brake (DB) action	Hold *2
5	Free-run (DB OFF)	Dynamic Brake (DB) action	Hold *2
6	Dynamic Brake (DB) action	Free-run (DB OFF)	Hold *2
7	Free-run (DB OFF)	Free-run (DB OFF)	Hold *2
8	Emergency stop *1	Dynamic Brake (DB) action	Clear *4
9	Emergency stop *1	Free-run (DB OFF)	Clear *4

*1 Emergency stop refers to a controlled immediate stop with servo-on.
The torque command value is limited during this process by Pr5.11 Emergency stop torque setup.

*2 If the positional command is kept applied or the motor is kept running with servo-off condition, positional deviation is accumulated, causing Err24.0 Excess positional deviation protection to be issued. If the servo is turned ON while the position or feedback scale is significantly deviating, the motor may rapidly operate to reduce the deviation to 0. Remember these requirements if you want to maintain the positional deviation/feedback scale deviation.

*3 Deceleration period is the time required for the running motor to speed down to 30 r/min. Once the motor speed drops below 30 r/min, it is treated as in stop state regardless of its speed.

*4 Positional deviation/feedback scale deviation is always cleared to 0.

Caution If an error occurs during servo-off, follow Pr5.10 Sequence at alarm. If the main power is turned off during servo-off, follow Pr5.07 Sequence during main power interruption.

Related page Refer to P.2-32, "Timing Chart"-Servo-ON/OFF action while the motor is at stall" of Preparation as well.

Pr5.07	Title	Sequence at main power OFF			Related control mode	P	S	T	F
	Range	0 to 9	Unit	—	Default	0			

Specify the status during deceleration after main power interrupt or after stoppage.
The relationship between the setup value of Pr5.06 and the operation and process at deviation counters is the same as that for Pr5.07 (sequence at main power OFF).

Caution If an error occurs with the main power supply turned off, Pr5.10 Sequence at alarm is applied to the operation.
When the main power supply is turned off with servo-on state, Err13.1 Main power undervoltage error occurs if Pr5.08 LV trip selection with main power off = 1, and the operation follows Pr5.10 Sequence at alarm.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4"

Pr5.08	Title	LV trip selection at main power OFF			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	1			

You can select whether or not to activate Err13 (Main power under-voltage protection) function while the main power shutoff continues for the setup of Pr5.09 (Main power-OFF detection time).

Setup value	Action of main power low voltage protection
0	When the main power is shut off during Servo-ON, Err13 will not be triggered and the driver turns to Servo-OFF. The driver returns to Servo-ON again after the main power resumption.
1	When the main power is shut off during Servo-ON, the driver will trip due to Err13 (Main power low voltage protection).

Caution This parameter is invalid when Pr5.09 (Detection time of main power OFF)=1000. Err13 (Main power under-voltage protection) is triggered when setup of P5.09 is long and P-N voltage of the main converter falls below the specified value before detecting the main power shutoff, regardless of the Pr5.08 setup.

Pr5.09 *	Title	Detection time of main power off			Related control mode	P	S	T	F
	Range	70 to 2000	Unit	1ms	Default	70			

You can set up the time to detect the shutoff while the main power is kept shut off continuously. The main power off detection is invalid when you set up this to 2000.

Pr5.10	Title	Sequence at alarm			Related control mode	P	S	T	F
	Range	0 to 7	Unit	—	Default	0			

Specify the status during deceleration and after stop, after occurrence of alarm.

Setup value	During deceleration *3	After stalling	Positional deviation/ feedback scale deviation
0	Dynamic Brake (DB) action	Dynamic Brake (DB) action	Clear *1
1	Free-run (DB OFF)	Dynamic Brake (DB) action	Clear *1
2	Dynamic Brake (DB) action	Free-run (DB OFF)	Clear *1
3	Free-run (DB OFF)	Free-run (DB OFF)	Clear *1
4	Action A: Emergency stop Action B: DB action *2	Dynamic Brake (DB) action	Clear *1
5	Action A: Emergency stop Action B: DB OFF *2	Dynamic Brake (DB) action	Clear *1
6	Action A: Emergency stop Action B: DB action *2	Free-run (DB OFF)	Clear *1
7	Action A: Emergency stop Action B: DB OFF *2	Free-run (DB OFF)	Clear *1

*1 Positional deviation/feedback scale deviation is maintained during alarm condition and will be cleared when the alarm is cancelled.

*2 Action of A/B: When an alarm requiring emergency stop occurs, the action A is selected when the setup value in the table is set within the range 4 to 7, causing emergency stop of operation. When an alarm not requiring emergency stop occurs, it triggers dynamic braking (DB) specified by action B, or free-running.

*3 Deceleration period is the time required for the running motor to speed down to 30 r/min.

Pr5.11	Title	Torque setup for emergency stop			Related control mode	P	S	T	F
	Range	0 to 500	Unit	%	Default	0			

Set up the torque limit at emergency stop.

Note When setup value is 0, the torque limit for normal operation is applied.

Pr5.12	Title	Over-load level setup			Related control mode	P	S	T	F
	Range	0 to 500	Unit	%	Default	0			

- You can set up the over-load level. The overload level becomes 115 [%] by setting up this to 0.
- Use this with 0 setup in normal operation. Set up other value only when you need to lower the over-load level.
- The setup value of this parameter is limited by 115[%] of the motor rating.

Pr5.13	Title	Over-speed level setup			Related control mode	P	S	T	F
	Default	0 to 20000	Unit	r/min	Default	0			

- If the motor speed exceeds this setup value, Err26.0 Over-speed protection occurs.
- The over-speed level becomes 1.2 times of the motor max. speed by setting up this to 0.

Pr5.14	Title	Motor working range setup			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1 revolution	Default	10			

- You can set up the movable range of the motor against the position command input range.
- When the motor movement exceeds the setup value, software limit protection of Err34.0 will be triggered.

Pr5.15 *	Title	I/F reading filter			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	0			

Select reading period of the control input signal.

Setup value	Reading period of the signal.
0	0.166ms
1	0.333ms
2	1ms
3	1.666ms

Exclude deviation counter clear input (CL) and command pulse inhibit input (INH).

Pr5.16 *	Title	Alarm clear input setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Select alarm clear input (A-CLR) recognition time.

Setup value	Recognition time
0	120ms
1	To Pr5.15 IF reading filter

Pr5.17	Title	Counter clear input mode			Related control mode	P S T F
	Range	0 to 4	Unit	—	Default	3

You can set up the clearing conditions of the counter clear input signal.

Setup value	Clear condition
0	Invalid
1	Clear at a level (no reading filter)
2	Clear at a level (with reading filter)
3	Clear at an edge (no reading filter)
4	Clear at an edge (with reading filter)

Note For signal width/timing requiring the deviation counter input, refer to P.3-38.

Pr5.18	Title	Invalidation of command pulse inhibit input			Related control mode	P S T F
	Range	0 to 1	Unit	—	Default	1

Select command pulse inhibit input enable/disable.

Setup value	INH input
0	Valid
1	Invalid

Pr5.19 *	Title	Command pulse inhibit input reading setup			Related control mode	P S T F
	Range	0 to 4	Unit	—	Default	0

Select command pulse inhibit input enable/disable signal reading period. When the status of several signals read during the predetermined reading period are same, update the signal status.

Setup value	Signal reading period
0	0.166ms
1	0.333ms
2	1ms
3	1.666ms
4	0.166 ms (no check for multiple coincidence)

Caution Longer reading period protects against operation error due to noise but decreases response to input signal.

Pr5.20 *	Title	Position setup unit select			Related control mode	P S T F
	Range	0 to 1	Unit	—	Default	0

Specify the unit to determine the range of positioning complete and excessive positional deviation.

Setup value	Unit
0	Command unit
1	Encoder unit

Note
- **For parameters which No. have a suffix of " * ",** changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4" • P.6-3 "Protective Function"

Pr5.21	Title	Selection of torque limit			Related control mode	P S T F
	Range	0 to 6	Unit	—	Default	1

You can set up the torque limiting method.

Setup value	Positive direction	Negative direction
0	P-ATL (0 to 10V)	N-ATL (–10 to 0V)
1	1st torque limit (Pr0.13)	
2	1st torque limit (Pr0.13)	2nd torque limit (Pr5.22)
3	TL-SEL OFF → 1st torque limit (Pr0.13) TL-SEL ON → 2nd torque limit (Pr5.22)	
4	P-ATL (0 to 10V)	N-ATL (0 to 10V)
5	P-ATL (0 to 10V)	
6	TL-SEL OFF	
	1st torque limit (Pr0.13)	2nd torque limit (Pr5.22)
	TL-SEL ON	
	External input positive direction torque limit (Pr5.25)	External input negative direction torque limit (Pr5.26)

Pr5.22	Title	2nd torque limit			Related control mode	P S T F
	Range	0 to 500	Unit	%	Default	500

You can set up the 2nd limit value of the motor output torque.
The value of parameter is limited to the maximum torque of the applicable motor.

Note For details of torque limit value, refer to P.2-49.

Pr5.23	Title	Torque limit switching setup 1			Related control mode	P S T F
	Range	0 to 4000	Unit	ms/100%	Default	0

Specify the rate of change (slope) from 1st to 2nd during torque limit switching.

Pr5.24	Title	Torque limit switching setup 2			Related control mode	P S T F
	Range	0 to 4000	Unit	ms/100%	Default	0

Specify the rate of change (slope) from 2nd to 1st during torque limit switching.

Pr5.25	Title	External input positive direction torque limit			Related control mode	P S T F
	Range	0 to 500	Unit	%	Default	500

Set up positive direction torque limit upon receiving TL-SEL with Pr5.21 Selection of torque limit set at 6.
The value of parameter is limited to the maximum torque of the applicable motor.

Note For details of torque limit value, refer to P.2-49.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.3-30... "Inputs and outputs on connector X4" • P.6-3 "Protective Function"

Pr5.26	Title	External input negative direction torque limit			Related control mode	P	S	T	F
	Range	0 to 500	Unit	%	Default	500			

Set up negative direction torque limit upon receiving TL-SEL with Pr5.21 Selection of torque limit set at 6.
The value of parameter is limited to the maximum torque of the applicable motor.

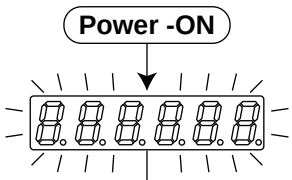
Note For details of torque limit value, refer to P.2-49.

Pr5.27	Title	Input gain of analog torque limit			Related control mode	P	S	T	F
	Range	10 to 100	Unit	0.1V/100%	Default	30			

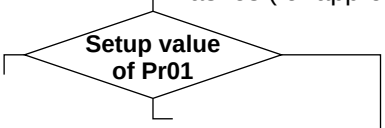
From the voltage [V] applied to the analog torque limit input (P-ATL, N-ATL), set conversion gain to torque limit [%].

Pr5.28 *	Title	LED initial status			Related control mode	P	S	T	F
	Range	0 to 35	Unit	—	Default	1			

You can select the type of data to be displayed on the front panel LED (7 segment) at the initial status after power-on.



Flashes (for approx. 2 sec) during initialization



Setup value	Content	Setup value	Content	Setup value	Content
0	Positional command deviation	12	Error factor and reference of history	24	Encoder positional deviation [Encoder unit]
1	Motor speed	13	Alarm Display	25	External scale deviation [External scale unit]
2	Positional command speed	14	Regenerative load factor	26	Hybrid deviation [Command unit]
3	Velocity control command	15	Over-load factor	27	Voltage across PN [V]
4	Torque command	16	Inertia ratio	28	Software version
5	Feedback pulse sum	17	Factor of no-motor running	29	Driver serial number
6	Command pulse sum	18	No. of changes in I/O signals	30	Motor serial number
8	External scale feedback pulse sum	20	Absolute encoder data	31	Accumulated operation time
9	Control mode	21	Absolute external scale position	32	Automatic motor recognizing function
10	I/O signal status	22	No. of encoder/ external scale communication errors monitor	33	Driver temperature
11	Analog input value	23	Communication axis address	35	Safety condition monitor

Related page For details of display, refer to P.2-58 "Setup of Parameter and Mode" of Preparation.

Pr5.29 *	Title	Baud rate setup of RS232 communication			Related control mode	P	S	T	F
	Range	0 to 6	Unit	—	Default	2			

You can set up the communication speed of RS232.

Note For baud rate setup value, refer to RS485 setup.

Pr5.30 *	Title	Baud rate setup of RS485 communication			Related control mode	P	S	T	F
	Range	0 to 6	Unit	—	Default	2			

You can set up the communication speed of RS485.

Setup value	Baud rate
0	2400bps
1	4800bps
2	9600bps
3	19200bps

Setup value	Baud rate
4	38400bps
5	57600bps
6	115200bps

Baud rate error is ±0.5% for 2400 to 38400 bps, and ±2% for 57,600 to 115,200 bps.

Pr5.31 *	Title	Axis address			Related control mode	P	S	T	F
	Range	0 to 127	Unit	—	Default	1			

During communication with the host (e.g. PC) to control multiple shafts, the shaft being accessed by the host should be identified.

Note When using RS232/RS485, the maximum valid value is 31.

Pr5.32 *	Title	Command pulse input maximum setup			Related control mode	P	S	T	F
	Range	250 to 4000	Unit	—	Default	4000			

Set the maximum number of pulses to be used as command pulse input. If the number of input pulses is outside the range, Err.27.0 Command pulse input frequency error protection occurs.

Caution The number of input pulses received by the driver is always checked. If the frequency of the received pulse is higher than the upper limit of the setting, input pulses are not accurately detected.

Pr5.33 *	Title	Pulse regenerative output limit setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Enable/disable detection of Err28.0 Pulse regenerative limit protection.

Setup value	Content
0	Invalid
1	Valid

Pr5.34	Title	For manufacturer's use			Related control mode	P	S	T	F
	Range	—	Unit	—	Default	4			

Fixed to 4.

Pr5.35 *	Title	Front panel lock setup			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Lock the operation on the front panel.

Setup value	Content
0	No limit on the front panel operation
1	Lock the operation on the front panel

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.2-49 "Setup of Torque Limit" • P.3-30... "Inputs and outputs on connector X4" • P.6-3 "Protective Function"

1. Details of parameter

[Class 6] Special setting

Pr6.00	Title	Analog torque feed forward conversion gain			Related control mode	P S T F
	Range	0 to 100	Unit	0.1V/100%	Default	0

- Set the input gain of analog torque feed forward.
  0 to 9 are invalid.

<Usage example of Analog torque feed forward>

- Setting bit 5 place of Pr6.10 Function expansion setup to 1 enables the analog torque feed forward. When the analog input 3 is used by another function (e.g. analog torque limit), the function becomes invalid.
- The voltage (V) applied to the analog input 3 is converted to the torque via Pr6.00 Analog torque feed forward conversion gain setup and added to the torque command (%).
- The conversion of analog input 3, input voltage [V], to the torque command [%] to the motor may be expressed mathematically as follows:

Torque command (%) = 100 × input voltage (V) / (Pr6.00 setup value × 0.1)

Pr6.02	Title	Velocity deviation excess setup			Related control mode	P S T F
	Range	0 to 100	Unit	r/min	Default	0

When the speed deviation (difference between internal positional command and actual speed) exceeds this value, Err24.1 Speed over deviation protection occurs.
This protection is not detected when the setup value is 0.

Pr6.04	Title	JOG trial run command speed			Related control mode	P S T F
	Range	0 to 500	Unit	r/min	Default	300

Set up the command speed used for JOG trial run (velocity control).

Related page Before using, refer to P.4-58 Preparation Trial Run.

Pr6.05	Title	Position 3rd gain valid time			Related control mode	P S T F
	Range	0 to 10000	Unit	0.1ms	Default	0

- Set up the time at which 3rd gain becomes valid.
- When not using this parameter, set Pr6.05 to 0 and Pr6.06 to 100.
- This is valid for only position control/full-closed control.

Pr6.06	Title	Position 3rd gain scale factor			Related control mode	P S T F
	Range	50 to 1000	Unit	%	Default	100

- Set up the 3rd gain by a multiplying factor of the 1st gain:
- 3rd gain = 1st gain × Pr6.06/100

Pr6.07	Title	Torque command additional value			Related control mode	P S T F
	Range	–100 to 100	Unit	%	Default	0

- Set up the offset load compensation value usually added to the torque command in a control mode except for the torque control mode.
- Update this parameter when the vertical axis mode for real time auto-tuning is valid.

Pr6.08	Title	Positive direction torque compensation value			Related control mode	P	S	T	F
	Range	–100 to 100	Unit	%	Default	0			

- Set up the dynamic friction compensation value to be added to the torque command when forward positional command is fed.
- Update this parameter when the friction compensation mode for real time auto-tuning is valid.

Pr6.09	Title	Negative direction torque compensation value			Related control mode	P	S	T	F
	Range	–100 to 100	Unit	%	Default	0			

- Set up the dynamic friction compensation value to be added to the torque command when negative direction positional command is fed.
- Update this parameter when the friction compensation mode for real time auto-tuning is valid.

Pr6.10	Title	Function expansion setup			Related control mode	P	S	T	F
	Range	0 to 63	Unit	—	Default	0			

Set up the function in unit of bit.

	Function	Setup value	
		0	1
bit 0	Speed observer	Invalid	Valid
bit 1	Disturbance observer	Invalid	Valid
bit 2	Disturbance observer operation setup	Always valid	Valid only when 1st gain is selected.
bit 3	Inertia ratio switching	Invalid	Valid
bit 4	Current response improvement	Invalid	Valid
bit 5	Analog torque FF	Invalid	Valid

* bit 0 = LSB

Pr6.11	Title	Current response setup			Related control mode	P	S	T	F
	Range	50 to 100	Unit	%	Default	100			

Fine tune the current response with respect to default setup (100%).

Pr6.13	Title	2nd Inertia ratio			Related control mode	P	S	T	F
	Range	0 to 10000	Unit	%	Default	250			

Set 1st inertia ratio.
You can set up the ratio of the load inertia against the rotor (of the motor) inertia.

Pr6.13 = (load inertia/ rotor inertia) × 100 [%]

Caution If the inertia ratio is correctly set, the setup unit of Pr1.01 and Pr1.06 becomes (Hz). When the inertia ratio of Pr0.04 is larger than the actual, the setup unit of the velocity loop gain becomes larger, and when the inertia ratio of Pr0.04 is smaller than the actual, the setup unit of the velocity loop gain becomes smaller.

Pr6.14	Title	Emergency stop time at alarm			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	1ms	Default	200			

Set up the time allowed to complete emergency stop in an alarm condition. Exceeding this time puts the system in alarm state.
When setup value is 0, immediate stop is disabled and the immediate alarm stop is enabled.

Pr6.15	Title	2nd over-speed level setup			Related control mode	P	S	T	F
	Range	0 to 20000	Unit	r/min	Default	0			

When the motor speed exceeds this setup time during emergency stop sequence in an alarm condition, Err26.1 2nd overspeed protection will be activated.
The over-speed level becomes 1.2 times of the motor max. speed by setting up this to 0.

Pr6.17 *	Title	Front panel parameter writing selection			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Specify the EEPROM writing procedure when parameter is edited form the front panel.

Setup value	Writing
0	Do not write to EEPROM at the same time
1	Write to EEPROM at the same time

Pr6.18 *	Title	Power-up wait time			Related control mode	P	S	T	F
	Range	0 to 100	Unit	0.1s	Default	0			

Set up the standard initialization time (1.5 s + α) after power-up.

Pr6.19 *	Title	Encoder Z phase setup			Related control mode	P	S	T	F
	Range	0 to 32767	Unit	pulse	Default	0			

If the number of output pulses per one motor revolution after division of pulse output is not an integer, fine adjust the width of encoder Z phase.

Pr6.20 *	Title	Z-phase setup of external scale			Related control mode	P	S	T	F
	Range	0 to 400	Unit	μs	Default	0			

Set up the Z phase regenerative width of feedback scale in unit of time. Even if the width of Z phase signal cannot be detected because the width equivalent of the travel distance from the feedback scale is too short, the Z phase signal will be output for at least the period set to this parameter.

Note • **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page • P.2-73 "EEPROM Writing" • P.3-30... "Inputs and outputs on connector X4" • P.6-3 "Protective Function"

Pr6.21 *	Title	Serial absolute external scale Z phase setup			Related control mode	P	S	T	F
	Range	0 to 2^{28}	Unit	pulse	Default	0			

Full-closed control using serial absolute feedback scale. When outputting pulses by using the feedback scale as the source of the output, set the Z phase output interval in units of A phase output pulses of the feedback scale (before multiplied by 4).

Setup value	Content
0	Output Z phase only at absolute 0 position of feedback scale.
1 to 268435456	After the power is fed to the driver, the Z phase, as it crosses the zero at the absolute position of feedback scale, is output in synchronous with the A phase. Subsequently, the Z phase is output at the A phase output pulse intervals set to this parameter.

Pr6.22 *	Title	A, B phase external scale pulse output method selection			Related control mode	P	S	T	F
	Range	0 to 1	Unit	—	Default	0			

Select the pulse regeneration method of A, B and Z parallel feedback scale.

Setup value	Regenerating method
0	Directly output the signals from A, B and Z parallel feedback scales.
1	Output A and B phase signals recovered from A, B and Z parallel feedback scales. Z-phase is output directly.

Pr6.23	Title	Disturbance torque compensating gain			Related control mode	P	S	T	F
	Range	–100 to 100	Unit	%	Default	0			

- Set up -100 to 100% compensating gain against disturbance torque.
- After setting up Pr6.24, increase Pr6.23.
  The disturbance suppressing capability increases by increasing the gain, but it is associated with increasing volume of operation noise.
  This means that well balanced setup can be obtained by adjusting Pr6.24 and Pr6.23.

Pr6.24	Title	Disturbance observer filter			Related control mode	P	S	T	F
	Range	0 to 2500	Unit	0.01ms	Default	53			

- Filter setup 10-2500, 0.01 ms. Set up the filter time constant according to the disturbance torque compensation.
- First, set up Pr6.24 to a larger value and check the operation with Pr6.23 Disturbance torque compensating gain set to a low value, and then gradually decrease the setup value of Pr6.24. A low filter setup value assures disturbance torque estimation with small delay and effectively suppresses effects of disturbance. However, this results in larger operation noise. Well balanced setup is required.

Pr6.27 *	Title	Alarm latch time selection			Related control mode	P	S	T	F
	Range	0 to 10	Unit	—	Default	5			

Set up the latch time.

Setup value	Content
0	Latch time: infinite
1 to 10	Latch time: 1 to 10 s

Pr6.31	Title	Real time auto tuning estimation speed			Related control mode	P	S	T	F
	Range	0 to 3	Unit	—	Default	1			

Set up the load characteristics estimation speed with the real time auto tuning being valid. A higher setup value assures faster response to a change in load characteristics but increases variations in disturbance estimation. Result of estimation is saved to EEPROM every 30 minutes.

Setup value	Mode	Description
0	No change	Stop estimation of load characteristics.
1	Almost constant	Response to changes in load characteristics in every minute.
2	Slower change	Response to changes in load characteristics in every second.
3 *	Faster change	Obtain best suitable estimation in response to changes in load characteristics.

* If the automatic oscillation detection is enabled by the support software PANATERM, the setup value 3 is used.

Pr6.32	Title	Real time auto tuning custom setup			Related control mode	P	S	T	F
	Range	–32768 to 32767	Unit	—	Default	0			

When the operation mode of real time auto tuning is set to the customize (Pr0.02 = 6), set the automatic adjusting function as shown below.

Bit	Content	Description
1 to 0	Load characteristics estimation *	Enable/disable the load characteristics estimation function. **Setup value** / **Function** **0** Disable 1 Enable * If the load characteristics estimation is disabled, the current setup cannot be changed even if the inertia ratio is updated according to the estimated value. When the torque compensation is updated by the estimated value, it is cleared to 0 (invalid).
3 to 2	Inertia ratio update	Set up update to be made based on result of the load characteristics estimation of Pr0.04 Inertia ratio. **Setup value** / **Function** **0** Use the current setup. 1 Update by the estimated value.

(continued)

Note
- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.
- The setup support software PANATERM can be downloaded from our web site.

Related page
- P.3-30... "Inputs and outputs on connector X4"

<table>
<tr>
<td>6 to 4</td>
<td>Torque compensation</td>
<td>Set up the update to be made according to the results of load characteristics estimation of Pr6.07 Torque command additional value, Pr6.08 positive direction torque compensation value and Pr6.09 negative direction torque compensation value.
<table>
<tr><th>Setup value</th><th>Function</th><th colspan="3">Compensation setup</th></tr>
<tr><td>0</td><td>Use current setup</td><td>Pr6.07</td><td>Pr6.08</td><td>Pr6.09</td></tr>
<tr><td>1</td><td>Disable torque compensation</td><td>0 clear</td><td>0 clear</td><td>0 clear</td></tr>
<tr><td>2</td><td>Vertical axis mode</td><td>Update</td><td>0 clear</td><td>0 clear</td></tr>
<tr><td>3</td><td>Friction compensation (low)</td><td>Update</td><td>Low</td><td>Low</td></tr>
<tr><td>4</td><td>Friction compensation (middle)</td><td>Update</td><td>Middle</td><td>Middle</td></tr>
<tr><td>5</td><td>Friction compensation (high)</td><td>Update</td><td>High</td><td>High</td></tr>
</table>
</td>
</tr>
<tr>
<td>7</td>
<td>Stiffness setup</td>
<td>Enable/disable the basic gain setup to be made according to Pr0.03 Real time auto tuning mechanical stiffness selection.
<table>
<tr><th>Setup value</th><th>Function</th></tr>
<tr><td>0</td><td>Disable</td></tr>
<tr><td>1</td><td>Enable</td></tr>
</table>
</td>
</tr>
<tr>
<td>8</td>
<td>Fixed parameter setup</td>
<td>Enable/disable the change of parameter that is normally set at a fixed value.
<table>
<tr><th>Setup value</th><th>Function</th></tr>
<tr><td>0</td><td>Use the current setup.</td></tr>
<tr><td>1</td><td>Set to a fixed value.</td></tr>
</table>
</td>
</tr>
<tr>
<td>10 to 9</td>
<td>Gain switching setup</td>
<td>Select the gain switching related parameter to be used when the real time auto tuning is enabled.
<table>
<tr><th>Setup value</th><th>Function</th></tr>
<tr><td>0</td><td>Use the current setup.</td></tr>
<tr><td>1</td><td>Disable gain switching.</td></tr>
<tr><td>2</td><td>Enable gain switching.</td></tr>
</table>
</td>
</tr>
</table>

Caution This parameter should be setup bit by bit. To prevent setting error, use of the setup support software is recommended when editing parameter.

<Setup procedure of bitwise parameter>

When setting parameter to a value other than 0, calculate the setup value of Pr6.32 in the following procedure.

1) Identify the LSB of the setup.
   Example: LSB of the torque compensation function is 4.
2) Multiply the setup value by power of 2 (LSB).
   Example: To set the torque compensation function to friction compensation (middle):
   $24 \times 4 = 64$.
3) Perform steps 1) and 2) for every setups, sum up the values which are to be Pr6.32 setup value.
   Example: Load characteristics measurement = enable, inertia ratio update = enable, torque compensation = friction compensation (middle), stiffness setup = enable, fixed parameter = set to a fixed value, gain switching setup = enable, then,
   $20 \times 1 + 22 \times 1 + 24 \times 4 + 27 \times 1 + 28 \times 1 + 29 \times 2 = 1477$

Pr6.34	Title	Hybrid vibration suppression gain			Related control mode	P	S	T	F
	Range	0 to 30000	Unit	0.1/s	Default	0			

Set up the hybrid vibration suppression gain for full-closed controlling.
First set it to the value identical to that of poison loop gain, and then fine tune as necessary.

Pr6.35	Title	Hybrid vibration suppression filter			Related control mode	P	S	T	F
	Range	0 to 6400	Unit	0.01ms	Default	10			

Set up the time constant of the hybrid vibration suppression filter for full-closed controlling.
While driving under full-closed control, gradually increase the setup value and check changes in the response.

Pr6.37	Title	Oscillation detecting level			Related control mode	P	S	T	F
	Range	0 to 1000	Unit	0.1%	Default	0			

Set up the oscillation detecting level.
Upon detection of a torque vibration whose level is higher than this setup value, the oscillation detection alarm will be issued.

Pr6.38 *	Title	Alarm mask setup			Related control mode	P	S	T	F
	Range	–32768 to 32767	Unit	—	Default	0			

Set up the alarm detection mask. Placing 1 to the corresponding bit position disables detection of the alarm condition.

Pr6.39	Title	For manufacturer's use			Related control mode	P	S	T	F
	Range	—	Unit	—	Default	0			

Fixed to 0.

Note

- **For parameters which No. have a suffix of " * "**, changed contents will be validated when you turn on the control power.

Related page

- P.3-30... "Inputs and outputs on connector X4"